ANALYTICALLY DENSE MORDELL–WEIL RANK JUMPS ON ELLIPTIC SURFACES VIA TRANSVERSE REALIZATION

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ABSTRACT. Let $\mathcal{W}_{\min,n}$ be the moduli stack of minimal Weierstrass fibrations of Faltings height n over an unparameterized $\mathbb{P}^1_{\mathbb{C}}$, and let $\mathcal{W}_n^{\mathrm{sb}}$ be its dense open substack of simply-branched surfaces with only I_1 singular fibres. For integers $n \geq 2$ and r satisfying

$$1 \leq r \leq \left\lfloor \frac{10n-2}{n-1} \right\rfloor$$

we prove that every point of $\mathcal{W}_n^{\mathrm{sb}}$ is a limit of pairwise non-isomorphic surfaces in this locus carrying r independent Mordell–Weil sections with indivisible Shioda classes, while the least canonical height of a non-zero section tends to infinity. On marked deformation charts, the moduli germ with these ordered sections identifies with a smooth transverse fixed-class Hodge locus germ of dimension $10n-2-r(n-1)$. Thus the rank-at-least-r locus is analytically dense in $\mathcal{W}_{\min,n}$.

The direct construction combines Shepherd–Barron's infinitesimal period calculation with a block Vandermonde minor, lattice approximation, and the holomorphic implicit function theorem, and uses neither monodromy nor o-minimality. An infinitesimal criterion also gives exact-rank-r density for $1 \leq r \leq 9$ when $n \geq 3$, and for $1 \leq r \leq 18$ when $n = 2$. Combining our construction with Baldi–Klingler–Ullmo's theorem, we obtain, for $n \geq 9$, smooth 8-dimensional local Hodge loci defined transversely by 10 independent classes in every non-empty analytic open subset, with Mordell–Weil rank exactly 10 outside an at most countable subset of each constructed Hodge locus and arbitrarily large uniform lower bounds for all non-zero section heights.

1. INTRODUCTION

The global sections of a Jacobian elliptic surface form its Mordell–Weil group. At each fixed Faltings height $n \geq 2$, a very general such surface over $\mathbb{P}^1_{\mathbb{C}}$ has rank zero. For $n \geq 3$, this follows from Cox's result on the very general Picard number [Cox90, Prop. 3.2]; the elliptic $K3$ case is classical. We study how additional sections appear under arbitrarily small deformations at the same height. We establish analytic density of Jacobian elliptic surfaces with Mordell–Weil rank at least r for every positive integer r in an explicit transverse range, and with rank exactly r for $1 \leq r \leq 9$ when $n \geq 3$ and $1 \leq r \leq 18$ when $n = 2$. Our construction coordinates local branch deformations through global period equations, producing independent sections whose canonical heights tend to infinity and transverse intersections of the elliptic surface period image with suitable simultaneous Hodge loci. After passage to marked local deformation charts, we identify the deformation germs of the constructed surfaces with their chosen ordered sections with the corresponding smooth Hodge locus germs.

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Throughout the paper we work over $\mathbb{C}$. By a Jacobian elliptic surface over $\mathbb{P}^1$ we mean a smooth projective surface S equipped with a morphism $\pi\colon S \to \mathbb{P}^1$ with connected fibres and smooth generic fibre of genus one, together with a distinguished section $\sigma\colon \mathbb{P}^1 \to S$, called the zero section. We assume that π is relatively-minimal, meaning that no fibre contains a (-1)-curve. Write $\omega_{S/\mathbb{P}^1}$ for the relative dualizing sheaf. Its fundamental line bundle and Faltings height are

$$\mathcal{L} = (R^1\pi_*\mathcal{O}_S)^\vee \simeq \pi_*\omega_{S/\mathbb{P}^1}, \qquad n = \deg \mathcal{L} = \chi(\mathcal{O}_S).$$

We assume $n \geq 2$. The minimal discriminant divisor has degree $12n$; see [Mir89, BHPVdV04, SS10] for background on elliptic surfaces and their invariants.

Let $\mathcal{W}_{\min,n}$ denote the moduli stack of minimal Weierstrass fibrations of height n over an unparameterized $\mathbb{P}^1$. It is obtained from the fixed-base height moduli stack of minimal Weierstrass fibrations with smooth generic fibre by quotienting by changes-of-coordinates on the base [PS25, Thms. 1.1–1.2]. Since the base $\mathbb{P}^1$ is unparameterized, isomorphisms may change its coordinates but must carry fibres to fibres and preserve the zero section. The fixed-base height moduli description by minimal weighted linear series is given in [BPS24, Prop. 7.5]. We identify the geometric points of $\mathcal{W}_{\min,n}$ with relatively-minimal Jacobian elliptic surfaces by taking the minimal resolution of each Weierstrass model. Analytic density is understood after pullback to analytic atlases; equivalently, it can be checked on the coarse moduli spaces with their analytic topology.

Write $E_S/\mathbb{C}(t)$ for the generic fibre and

$$\mathrm{MW}(S/\mathbb{P}^1) = E_S(\mathbb{C}(t))$$

for the Mordell–Weil group of sections. Let F and O denote the fibre and zero-section classes, and let $R(S)$ be the root lattice generated by the components of reducible fibres that do not meet O. Let $\mathrm{NS}(S)$ denote the Néron–Severi group of S. The trivial lattice $\mathrm{Triv}(S) \subset \mathrm{NS}(S)$ is generated by F, O, and these components. The Shioda–Tate description [Shi90, Thm. 1.3] gives

$$0 \longrightarrow \mathrm{Triv}(S) \longrightarrow \mathrm{NS}(S) \longrightarrow \mathrm{MW}(S/\mathbb{P}^1) \longrightarrow 0.$$

Writing $\rho(S) = \mathrm{rk}\,\mathrm{NS}(S)$, we obtain

$$\rho(S) = 2 + \mathrm{rk}\,R(S) + \mathrm{rk}\,\mathrm{MW}(S/\mathbb{P}^1). \tag{1}$$

The root-lattice term is determined by the Kodaira types of the singular fibres. To remove this contribution, we work on the squarefree-discriminant locus $\mathcal{W}_n^{\mathrm{I}_1} \subset \mathcal{W}_{\min,n}$. Every singular fibre on this locus is of Kodaira type I_1, that is, an irreducible rational curve with a single ordinary node. Thus $R(S) = 0$, and (1) becomes

$$\mathrm{rk}\,\mathrm{MW}(S/\mathbb{P}^1) = \rho(S) - 2.$$

On this locus, algebraic classes that are linearly independent modulo $\langle F, O\rangle$ give linearly independent Mordell–Weil sections.

To describe these classes, put

$$\Lambda_n = \langle F, O\rangle^\perp \subset H^2(S,\mathbb{Z})/\mathrm{tors}.$$

Its isomorphism type depends only on n; we use the same symbol for a fixed abstract model. Throughout the paper, integral cohomology lattices are understood

modulo torsion. For a lattice or finitely generated abelian group A, write $A_{\mathbf{K}} = A \otimes_{\mathbb{Z}} \mathbf{K}$ for $\mathbf{K} \in \{\mathbb{Q}, \mathbb{R}, \mathbb{C}\}$. Our primitive Hodge-space conventions are

$$H^{1,1}_{\mathrm{prim}}(S) = H^{1,1}(S) \cap \Lambda_{n,\mathbb{C}},$$
$$H^{1,1}_{\mathrm{prim}}(S, \mathbb{R}) = H^{1,1}(S) \cap \Lambda_{n,\mathbb{R}}.$$

We use the half-normalized Néron–Tate pairing

$$\langle P, Q \rangle_{\widehat{h}} = \frac{1}{2} \langle P, Q \rangle_{\mathrm{Sh}}, \qquad \widehat{h}(P) = \langle P, P \rangle_{\widehat{h}},$$

where $\langle\, , \rangle_{\mathrm{Sh}}$ is Shioda's lattice height pairing. On the all-I_1 locus, a non-zero section P satisfies $\widehat{h}(P) = n + P \cdot O$, where $P \cdot O$ counts its intersections with the zero section, with multiplicity; see Proposition 3.14. On this locus, we call the orthogonal projection of the section class $[P]$ to Λ_n its *Shioda class*; explicitly, $\gamma_P = [P] - [O] - (n + P \cdot O)[F] \in \Lambda_n$. The Shioda map identifies the Mordell–Weil group with the primitive integral Hodge lattice on the all-I_1 locus:

$$\sigma_{\mathrm{Sh}} \colon \mathrm{MW}(S/\mathbb{P}^1) \xrightarrow{\sim} \Lambda_n \cap H^{1,1}(S), \qquad P \longmapsto \gamma_P.$$

For this integral identification, see the proof of Theorem 3.11.

The moduli stack $\mathcal{W}_n^{\mathrm{I}_1}$ is irreducible of dimension $10n - 2$, as follows from the quotient description in [PS25, Thms. 1.1 and 1.2(a)]. Each Jacobian elliptic surface in this locus has a classifying morphism [HP19, Prop. 9]

$$\varphi_S \colon \mathbb{P}^1 \longrightarrow \overline{\mathcal{M}}_{1,1}.$$

Here $\overline{\mathcal{M}}_{1,1}$ is the moduli stack of stable genus one curves with one marked point, and $\mathcal{M}_{1,1}$ is its open substack parametrizing smooth curves. This morphism is unramified over the boundary $\overline{\mathcal{M}}_{1,1} \smallsetminus \mathcal{M}_{1,1}$; see the local calculation in the proof of [HP19, Prop. 11]. Its stack-theoretic ramification divisor $\mathrm{Ram}(\varphi_S)$ is the zero divisor of its differential; it differs from the ramification divisor of the coarse j-map. We call S *simply-branched* when $\mathrm{Ram}(\varphi_S)$ is reduced, and denote the resulting substack by

$$\mathcal{W}_n^{\mathrm{sb}} \subset \mathcal{W}_n^{\mathrm{I}_1}.$$

Distinct ramification points are allowed to have the same image under φ_S. Lemmas 3.2 and 3.3 show that $\mathcal{W}_n^{\mathrm{sb}}$ is Zariski open and dense in $\mathcal{W}_{\min,n}$, hence also analytically dense.

Elliptic $K3$ surfaces, which constitute the Calabi–Yau case $n = 2$, provide the archetype for our question. Since $F \cdot F = 0$, $F \cdot O = 1$, and $O^2 = -2$, the classes $F, O + F$ identify $\langle F, O \rangle$ with the hyperbolic plane U. We recall a standard consequence of local Torelli and the Shioda–Tate formula, including the proof to explain the comparison with higher height $n \geq 3$.

Proposition 1.1. *The locus*

$$\left\{ [S] \in \mathcal{W}_2^{\mathrm{I}_1} : \mathrm{rk}\,\mathrm{MW}(S/\mathbb{P}^1) = 18 \right\}$$

is analytically dense in $\mathcal{W}_2^{\mathrm{I}_1}$. Consequently, for every $1 \leq r \leq 18$, the locus of Jacobian elliptic surfaces with Mordell–Weil rank at least r is analytically dense in $\mathcal{W}_2^{\mathrm{I}_1}$.

Proof. Fix $[S] \in \mathcal{W}_2^{\mathrm{I}_1}$. The lattice $\Lambda_2 = \langle F, O\rangle^\perp$ has signature $(2,18)$. By local Torelli, the marked deformations of S on which F and O remain algebraic are locally parametrized by an open subset of a connected component $\mathcal{D}_2^+$ of

$$\big\{\Pi \subset \Lambda_{2,\mathbb{R}} : \Pi \text{ is an oriented positive two-plane}\big\};$$

see [Huy16, Ch. 6, Prop. 2.8 and §2.4]. The line bundles $\mathcal{O}_S(F)$ and $\mathcal{O}_S(O)$ extend over this fixed-class deformation space. On the central surface, their spaces of sections have dimensions two and one, respectively, and their first cohomology groups vanish. Cohomology and base change therefore extend the pencil and its zero-section divisor. After shrinking, the pencil remains basepoint free and the divisor remains a smooth rational curve; their intersection number one makes this divisor a section. The squarefree discriminant condition is open, so we obtain a marked chart B in $\mathcal{W}_2^{\mathrm{I}_1}$ whose period image is open in $\mathcal{D}_2^+$.

Positive two planes defined over $\mathbb{Q}$ are dense in $\mathcal{D}_2^+$. Indeed, an oriented basis of any positive real two-plane can be approximated by vectors in $\Lambda_{2,\mathbb{Q}}$, and sufficiently close approximations still span positive planes with the induced orientation. Choose a rational plane Π in the period image of B, and let S_Π be the corresponding surface.

Since Π is defined over $\mathbb{Q}$, its orthogonal complement is a rational subspace of dimension 18. Hence

$$\mathrm{rk}\big(\Pi^\perp \cap \Lambda_2\big) = 18.$$

These integral classes are of type $(1,1)$ on S_Π, so the Lefschetz $(1,1)$-theorem makes them algebraic. Together with F and O, these give twenty independent classes in $\mathrm{NS}(S_\Pi)$. Since divisor classes lie in $H^{1,1}(S_\Pi)$, we have

$$20 \le \rho(S_\Pi) \le h^{1,1}(S_\Pi) = 20.$$

Hence $\rho(S_\Pi) = 20$.

The surface S_Π lies in $\mathcal{W}_2^{\mathrm{I}_1}$, so its singular fibres are irreducible. Thus (1) gives

$$\mathrm{rk}\,\mathrm{MW}(S_\Pi/\mathbb{P}^1) = \rho(S_\Pi) - 2 = 18.$$

Rational positive planes occur in every open subset of the local period image, proving density. The assertion for rank at least r follows immediately. ∎

This is the classical period-theoretic density argument with the fibre and zero-section classes fixed. For related results, Oguiso proves density of every prescribed Picard number in the full Kuranishi space and constructs a hierarchy of narrow Mordell–Weil lattices of Jacobian $K3$ surfaces [Ogu03, Cor. 1.3 and Thm. 1.4].

In the $K3$ case, local Torelli identifies the marked deformation space preserving F and O with an open subset of $\mathcal{D}_2^+$. At higher height, the dimension of the ambient period domain prevents such local openness. The primitive Hodge numbers are

$$h^{2,0} = h^{0,2} = n-1, \qquad h^{1,1}_{\mathrm{prim}} = 10n-2.$$

For general n, let $\mathcal{D}_n^+$ denote the connected component of the corresponding polarized weight-2 period domain containing the marked periods under consideration. For $n \ge 3$, its dimension satisfies

$$\dim_{\mathbb{C}} \mathcal{D}_n^+ = (n-1)(10n-2) + \binom{n-1}{2} > 10n-2 = \dim \mathcal{W}_n^{\mathrm{I}_1}.$$

The local elliptic surface period image therefore has positive codimension in $\mathcal{D}_n^+$. Density of Hodge loci in the ambient domain alone no longer guarantees that they meet this image.

A fixed class $\gamma \in \Lambda_n$ with $\gamma^2 < 0$ defines a Hodge locus of complex codimension $n-1$ in $\mathcal{D}_n^+$. After restriction to the elliptic surface deformation space, however, its defining equations need not remain independent. We must establish the required intersections using deformations of Jacobian elliptic surfaces themselves.

Our main theorem realizes, near every simply-branched surface, r independent Mordell–Weil sections through transverse period conditions, with the least canonical height of a non-zero section tending to infinity while the Faltings height $n \geq 2$ of the surfaces remains fixed.

Main Theorem 1.2. *Let $n \geq 2$ and $r \geq 1$ be integers satisfying*

$$r(n-1) \leq 10n-2.$$

Let $\mathcal{W}_{\min,n}$ denote the complex moduli stack of minimal Weierstrass fibrations of Faltings height n over an unparameterized $\mathbb{P}^1_{\mathbb{C}}$. Let $\mathcal{W}_n^{\mathrm{sb}} \subset \mathcal{W}_{\min,n}$ be the locus where every singular fibre is of Kodaira type I_1 and the differential of the classifying morphism

$$\varphi_S \colon \mathbb{P}^1_{\mathbb{C}} \longrightarrow \overline{\mathcal{M}}_{1,1}$$

has reduced zero divisor. For every $[S] \in \mathcal{W}_n^{\mathrm{sb}}$, there are pairwise non-isomorphic Jacobian elliptic surfaces

$$[S_m] \in \mathcal{W}_n^{\mathrm{sb}}, \qquad [S_m] \longrightarrow [S]$$

in the analytic topology, and independent Mordell–Weil sections $P_{1,m}, \ldots, P_{r,m} \in \mathrm{MW}(S_m/\mathbb{P}^1)$, with the following properties.

(1) *The Shioda classes $\gamma_{a,m} \in \Lambda_n$ of the sections $P_{a,m}$ are indivisible, and*

$$\min_{0 \neq P \in \mathrm{MW}(S_m/\mathbb{P}^1)} \widehat{h}(P) \longrightarrow \infty \qquad \text{as } m \to \infty.$$

(2) *On a marked local deformation chart, the $r(n-1)$ period equations under which these classes remain of type $(1,1)$ have linearly independent differentials at S_m. Their common zero locus is therefore smooth of complex codimension $r(n-1)$ near S_m.*

(3) *The moduli stack of surfaces in $\mathcal{W}_n^{\mathrm{sb}}$ with r ordered non-zero sections is smooth at $(S_m, P_{1,m}, \ldots, P_{r,m})$, of complex dimension $10n-2-r(n-1)$. After base change to a marked deformation chart, the forgetful morphism identifies its germ at the chosen lift with the germ of the locus in* (2).

Consequently, every analytic neighbourhood of every point of $\mathcal{W}_{\min,n}$ contains infinitely many pairwise non-isomorphic surfaces in $\mathcal{W}_n^{\mathrm{sb}}$ carrying r independent Mordell–Weil sections and satisfying $\widehat{h}(P) > H$ for every non-zero Mordell–Weil section P, for any prescribed $H > 0$. In particular, the locus of surfaces in $\mathcal{W}_n^{\mathrm{sb}}$ with Mordell–Weil rank at least r is analytically dense in $\mathcal{W}_{\min,n}$.

For every $n \geq 2$, taking $r = 10$ gives smooth 8-dimensional simultaneous Hodge germs in marked deformation charts of $\mathcal{W}_n^{\mathrm{sb}}$. Along each germ, the 10 chosen independent sections persist with constant height-pairing matrix. For every non-empty analytic open subset $U \subset \mathcal{W}_{\min,n}$ and every $H > 0$, one can choose such a germ with

a representative Z whose image lies in U and such that every non-zero Mordell–Weil section of every surface parametrized by Z has canonical height greater than H.

The next statement makes (1)–(3) precise in terms of a marked deformation chart.

Theorem 1.3. *In the situation of Theorem 1.2, the surfaces S_m and the sections $P_{a,m}$ may be chosen as follows. The surfaces occur in a simply-connected analytic étale local deformation chart $B \to \mathcal{W}_n^{\mathrm{sb}}$ of dimension $10n-2$, with $S_{b_0} = S$, $S_{b_m} = S_m$, and $b_m \to b_0$. A marking identifies the lattices $\langle F_b, O_b\rangle^{\perp} \subset H^2(S_b,\mathbb{Z})/\mathrm{tors}$ with a fixed lattice Λ_n by parallel transport preserving the cup-product pairing, where F_b, O_b are the fibre and zero-section classes. Under this marking, for each m the Shioda classes $\gamma_{1,m},\dots,\gamma_{r,m} \in \Lambda_n$ of $P_{1,m},\dots,P_{r,m}$ are linearly independent, indivisible, and of type $(1,1)$ on S_m. Their canonical heights satisfy*

$$\widehat{h}(P_{a,m}) = -\frac{1}{2}\gamma_{a,m}^2 \longrightarrow \infty \qquad (1 \le a \le r).$$

After shrinking B and discarding finitely many initial terms, choose a holomorphic frame $\omega_0(b),\dots,\omega_{n-2}(b)$ of the Hodge bundle F^2 over B, viewed in $\Lambda_n \otimes_{\mathbb{Z}} \mathbb{C}$ through the marking. With a dot denoting the cup-product pairing, define

$$\Phi_m \colon B \longrightarrow \mathbb{C}^{r(n-1)}, \qquad \Phi_m(b) = \big(\gamma_{a,m}\cdot\omega_j(b)\big)_{\substack{1\le a\le r\\ 0\le j\le n-2}}.$$

Then

$$\Phi_m(b_m) = 0, \qquad \mathrm{rk}_{\mathbb{C}}(d\Phi_m)_{b_m} = r(n-1).$$

Thus the simultaneous fixed-class Hodge locus $\Phi_m^{-1}(0)$ is smooth of complex codimension $r(n-1)$ near b_m. The moduli stack of Jacobian elliptic surfaces in $\mathcal{W}_n^{\mathrm{sb}}$ equipped with r ordered non-zero sections is smooth at $(S_m, P_{1,m},\dots,P_{r,m})$, of complex dimension $10n-2-r(n-1)$. After base change to B, forgetting the additional sections identifies its germ at the chosen lift over b_m with $(\Phi_m^{-1}(0), b_m)$.

At a simply-branched surface of height n, the $10n-2$ ramification points supply independent deformation directions, while each class contributes $n-1$ period equations. Transverse realization of r classes therefore requires

$$r(n-1) \le 10n-2 \quad\Longleftrightarrow\quad r \le 10 + \left\lfloor \frac{8}{n-1} \right\rfloor.$$

Equivalently, this bound requires the expected dimension $10n-2-r(n-1)$ of the simultaneous fixed-class Hodge locus to be non-negative. Thus the transverse range stabilizes at $r \le 10$ for $n \ge 10$. Beyond this range, the expected dimension is negative, so every non-empty intersection of the local period image of $\mathcal{W}_n^{\mathrm{sb}}$ with the corresponding simultaneous fixed-class Hodge locus has dimension greater than expected and is therefore atypical. In particular, every such realization is non-transverse.

The constant dimension 8 also has a moduli-theoretic interpretation: 10 additional sections together with the zero section give 11 markings, and the corresponding 11-dimensional fibre-product target has trivial canonical bundle on the locus containing the section maps. Remark 3.17 explains the expected deformation

dimension $11-3=8$ for these maps from $\mathbb{P}^1$, after quotienting by source automorphisms. It also compares this threshold with the transition from $\kappa(\overline{M}_{1,k})=-\infty$ for $1\le k\le 10$ to $\kappa(\overline{M}_{1,11})=0$.

Schiffer variations arise from elementary local changes of the holomorphic gluing of a Riemann surface [Sch53, §3]. On the simply-branched locus, Shepherd–Barron's adaptation to elliptic surfaces provides local deformation coordinates indexed by the ramification points of the classifying morphism [SB23, Prop. 4.4]. Their tangent directions form a basis of the deformation tangent space, so every first-order deformation is a linear combination of these elementary local variations. Each direction induces a rank-one map from $H^{2,0}(S)$ to $H^{1,1}_{\mathrm{prim}}(S)$ [SB23, Thm. 1.4(3)]. We call non-zero generators of these image lines the *Schiffer image classes*; they form an orthogonal basis of primitive $(1,1)$-cohomology [SB23, Thm. 4.11(4)]. Evaluation of holomorphic two-forms at distinct ramification points gives Vandermonde minors. Our block construction uses these to show that a maximal Jacobian minor of the simultaneous period equations is not identically zero.

We choose a real tuple in $H^{1,1}_{\mathrm{prim}}(S,\mathbb{R})$ with negative-definite Gram matrix at which the chosen minor does not vanish. We approximate its directions by those of indivisible lattice vectors whose norms tend to infinity in a fixed auxiliary Euclidean norm. Applying the holomorphic implicit function theorem, with the normalized lattice vectors as parameters, gives nearby surfaces on which all their period pairings vanish. Rescaling to the original integral vectors preserves these vanishing pairings. Since the vectors are real, they are also orthogonal to $H^{0,2}$ and hence have type $(1,1)$.

Independence persists under sufficiently close approximation. Moreover, the normalized limiting vectors have negative square, so $-\gamma_{a,m}^2\to+\infty$ for every a. The Lefschetz $(1,1)$-theorem makes these classes algebraic, and on the all-I_1 locus Shioda's identification turns them into independent sections satisfying

$$\widehat{h}(P_{a,m})=-\frac{1}{2}\gamma_{a,m}^2\longrightarrow+\infty.$$

These heights are integers because Λ_n is even. To control the heights of all non-zero sections, we use Lemma 3.7. On a relatively compact marked chart, only finitely many non-zero integral classes can become Hodge with $-\gamma^2/2\le H$. Their proper closed Hodge loci have an open dense complement. Applying the local realization theorem in these complements, with increasing H and shrinking neighbourhoods, makes the least canonical height of a non-zero section tend to infinity.

To obtain exact rank one, we use an explicit infinitesimal criterion. Suppose that a non-zero class $\alpha\in\Lambda_n$ is indivisible, is of type $(1,1)$ at a point of $\mathcal{W}_n^{\mathrm{sb}}$, and pairs non-trivially with every Schiffer image class. For each $\beta\in\Lambda_n\smallsetminus\mathbb{Q}\alpha$ that is Hodge at that point, some first-order deformation preserving the Hodge type of α does not preserve that of β. Consequently, outside a countable union of proper closed analytic subsets, a sufficiently small connected representative of the one-class Hodge germ parametrizes surfaces of Mordell–Weil rank exactly one; see Proposition 3.19. Combining this criterion and its simultaneous version with rational approximation and local realization gives the following result, proved in Corollary 3.23.

Corollary 1.4. *Let $n \geq 2$ and $r \geq 1$ be integers, with $r \leq 18$ if $n = 2$ and $r \leq 9$ if $n \geq 3$. Then the locus of surfaces in $\mathcal{W}_n^{\mathrm{sb}}$ with Mordell–Weil rank exactly r is analytically dense in $\mathcal{W}_{\min,n}$.*

The direct proof of Corollary 1.4 makes classical infinitesimal Hodge-theoretic criteria explicit in Schiffer deformation coordinates and uses neither monodromy nor o-minimality; see Subsection 3.7. Combining our realization theorem with Baldi–Klingler–Ullmo's algebraicity theorem [BKU24, Thm. 3.7] gives further control of higher-rank points. For $n \geq 9$, the 8-dimensional 10-class germs can be chosen in every non-empty analytic open subset so that the Mordell–Weil rank is exactly 10 outside an at most countable subset of each constructed Hodge locus. Their representatives may simultaneously be chosen so that every non-zero section on every parametrized surface has canonical height greater than any prescribed bound; see Corollary 3.26.

Cox and Kloosterman studied the Noether–Lefschetz and Mordell–Weil rank loci of Jacobian elliptic surfaces. For $n \geq 3$, Cox proves that the positive Mordell–Weil rank locus has dimension $9n - 1$ [Cox90] and that the union of the codimension-$(n-1)$ components of the Noether–Lefschetz locus is dense in the classical topology [Cox90, Thm. 3.1(iii)]. For every $n \geq 2$ and every $H > 0$, he also proves density of surfaces with positive Mordell–Weil rank for which every non-zero section P satisfies $P \cdot O \geq H$ [Cox90, Thm. 4.4]. On the all-I_1 locus, our normalization satisfies $\widehat{h}(P) = n + P \cdot O$, so this also gives arbitrarily large lower bounds for canonical heights. Kloosterman determines the dimensions of higher Noether–Lefschetz loci on the nonconstant-j locus and obtains dimension bounds for Mordell–Weil rank loci there [Klo07, Thm. 1.1 and Cor. 1.3].

These results fit into the general study of density and distribution of Hodge loci. Under suitable hypotheses on a weight-two variation, Tayou–Tholozan prove equidistribution of transverse refined Noether–Lefschetz points, ordered by discriminant, when the expected dimension is zero [TT23, Thm. 6.10]. Our proof gives a direct local construction of simultaneous transverse Hodge loci in $\mathcal{W}_n^{\mathrm{sb}}$ through branch deformations, lattice approximation, and the holomorphic implicit function theorem. It produces ordered tuples of independent sections with indivisible Shioda classes and canonical heights tending to infinity. At each constructed tuple, we also identify the local moduli of surfaces with the chosen sections: after base change to a marked local deformation chart, the forgetful morphism identifies the section-moduli germ with the corresponding smooth simultaneous fixed-class Hodge locus germ.

Remark 1.5. Higher Mordell–Weil ranks occur in classical Delsarte examples with cyclic symmetry; see [Hei12] for a classification and rank computations. Shioda's isotrivial example has rank 68 at height 60 [Shi92, pp. 109–110]. The non-isotrivial example studied by Shioda [Shi86] and Stiller [Sti87, Thm. 4.2] has rank 56 at height 210. The former has only type-II singular fibres. The latter admits the model

$$y^2 = x^3 + x + t^{1260},$$

whose singular fibres are all of type I_1, but whose stack-theoretic ramification divisor is

$$\mathrm{Ram}(\varphi_S) = 1259[0] + 839[\infty].$$

Thus neither example lies in $\mathcal{W}_n^{\mathrm{sb}}$. Nevertheless, these classical examples can be approximated, at fixed Faltings height, by surfaces in our controlled rank-10 families; see Corollary 3.26.

The natural question of whether Mordell–Weil ranks of positive-height Jacobian elliptic surfaces over $\mathbb{P}^1_{\mathbb{C}}$ are unbounded as the height grows remains open [SS10, Question 14.3]. Over each fixed finite field $\mathbb{F}_q$, Mordell–Weil ranks over $\mathbb{F}_q(t)$ are unbounded: Tate–Shafarevich established this for isotrivial curves in odd characteristic [Tv67], and Ulmer constructed non-isotrivial examples in every positive characteristic [Ulm02, Thm. 1.5].

1.1. **Outline of the paper.** Section 2 records the primitive lattice and the period conditions used in the proof. Section 3 establishes simultaneous transverse realization, the height bounds, and analytic density. Subsection 3.6 identifies canonical heights with two-pointed degrees and proves the section-moduli smoothness and germ comparison asserted in Theorem 1.3, completing the proofs of the main realization theorems. Subsection 3.7 establishes exact-rank density in the ranges of Corollary 3.23. Finally, Subsection 3.8 applies Baldi–Klingler–Ullmo's algebraicity theorem using Lönne's monodromy computation and establishes the rank-10 consequence.

2. Primitive cohomology and period conditions

We recall the lattice and Hodge-theoretic facts used in the realization argument. Our general reference for Hodge theory, period domains, and variations of Hodge structure throughout the paper is Carlson–Müller-Stach–Peters [CMSP17]. We use a dot for the intersection pairing and its complex-bilinear extension.

2.1. **The primitive lattice.** Let S be a smooth complex projective surface, and let $\pi\colon S \to \mathbb{P}^1$ be a relatively-minimal Jacobian elliptic fibration with $\chi(\mathcal{O}_S) = n \geq 2$. Let F and O denote a fibre divisor and the zero section, respectively; we use the same symbols for their divisor classes. The canonical-bundle formula gives $K_S = (n-2)F$. Since $F \cdot F = 0$ and $F \cdot O = 1$, adjunction gives $O^2 = -n$. The standard invariant formulas

$$q(S) = 0, \qquad p_g(S) = n-1, \qquad e(S) = 12n$$

give $b_2(S) = 12n-2$; see [Mir89, SS10]. Hodge decomposition and the Hodge index theorem give intersection signature $(2n-1, 10n-1)$ on $H^2(S,\mathbb{Z})/\mathrm{tors}$.

The fibre and zero-section classes have intersection matrix

$$\begin{pmatrix} 0 & 1 \\ 1 & -n \end{pmatrix}.$$

Its determinant is -1, so $\langle F, O\rangle$ is a primitive unimodular sublattice of signature $(1,1)$. Its orthogonal complement

$$\Lambda_n = \langle F, O\rangle^{\perp} \subset H^2(S,\mathbb{Z})/\mathrm{tors}$$

therefore has rank $12n-4$ and signature $(2n-2, 10n-2)$. Orthogonal projection gives an integral decomposition

$$H^2(S,\mathbb{Z})/\mathrm{tors} = \langle F, O\rangle \oplus \Lambda_n.$$

We recall the standard description of the primitive lattice and its proof [Gre20, Thm. 24]. The lattice Λ_n is even. Indeed, the Wu formula gives

$$\gamma^2 \equiv \gamma \cdot K_S = (n-2)\gamma \cdot F = 0 \pmod 2 \qquad (\gamma \in \Lambda_n).$$

Poincaré duality makes $H^2(S,\mathbb{Z})/\mathrm{tors}$ unimodular, so the orthogonal decomposition also shows that Λ_n is unimodular. The classification of indefinite even unimodular lattices gives

$$\Lambda_n \simeq U^{\oplus(2n-2)} \oplus E_8(-1)^{\oplus n}. \tag{2}$$

Here U is the even hyperbolic plane, with Gram matrix $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, and $E_8(-1)$ is the negative-definite E_8 lattice; see [Ser73, Ch. V, Thm. 5]. We use Λ_n also for a fixed abstract model of this lattice.

The primitive Hodge numbers are $h^{2,0} = h^{0,2} = p_g(S) = n-1$ and $h^{1,1}_{\mathrm{prim}} = (12n-4) - 2(n-1) = 10n-2$. The intersection pairing is positive definite on the real subspace

$$\left(H^{2,0}(S) \oplus H^{0,2}(S)\right) \cap \Lambda_{n,\mathbb{R}}$$

and negative definite on $H^{1,1}_{\mathrm{prim}}(S,\mathbb{R})$. In particular, every non-zero class $\gamma \in \Lambda_n \cap H^{1,1}(S)$ satisfies

$$-\frac{1}{2}\gamma^2 \in \mathbb{Z}_{>0}.$$

Here primitive cohomology means the orthogonal complement of F and O. To avoid a clash with this usage, we call a lattice vector *indivisible* if it is non-zero and not of the form $k\gamma'$ with γ' in the lattice and $k \geq 2$ an integer.

2.2. **Periods and fixed-class Hodge loci.** A primitive Hodge structure is determined by its space of holomorphic two-forms. Let $\mathcal{D}_n^+$ be a connected component of the period domain parametrizing $(n-1)$-dimensional subspaces $F^2 \subset \Lambda_{n,\mathbb{C}}$ satisfying

$$F^2 \cdot F^2 = 0, \qquad \omega \cdot \overline{\omega} > 0 \quad \text{for every } 0 \neq \omega \in F^2.$$

Its complex dimension is $(n-1)(10n-2) + \binom{n-1}{2}$. It is a type-IV Hermitian symmetric domain when $n = 2$; for $n \geq 3$, it is a Griffiths period domain that is not Hermitian symmetric. For this period-domain description in the elliptic-surface setting, see [Gre20, §3, p. 2345]; for the general Hodge-theoretic background, see [CMSP17, Ch. 4].

For a real class $\gamma \in \Lambda_{n,\mathbb{R}}$, the condition

$$\gamma \perp F^2$$

is equivalent to being of type $(1,1)$. Indeed, reality also gives orthogonality to $\overline{F^2}$, and the orthogonal complement of $F^2 \oplus \overline{F^2}$ is the primitive $(1,1)$-subspace.

If $\gamma^2 < 0$, its fixed-class Hodge locus in $\mathcal{D}_n^+$ is smooth of complex codimension $n-1$; see [Gre20, §3, p. 2345] and [Gar18, §2.4]. Requiring $F^2 \subset \gamma^\perp$ removes one negative direction from the intersection space, replacing $10n-2$ by $10n-3$ in the dimension formula above. More generally, a fixed negative-definite span

of dimension r defines a smooth ambient Hodge locus of complex codimension $r(n-1)$.

To express these conditions on a family, let B be a small simply-connected chart carrying a smooth projective family of height n Jacobian elliptic surfaces with zero section. The fibre and zero-section classes remain flat and algebraic. Choose a marking at a base point $b_0 \in B$, that is, an isometry from the primitive cohomology lattice of S_{b_0} to Λ_n. Since B is simply-connected, parallel transport extends this to compatible identifications of the primitive cohomology lattices of all fibres with Λ_n. Choose a local holomorphic frame $\omega_0(b), \dots, \omega_{n-2}(b)$ of the Hodge bundle F^2. For fixed classes $\gamma_1, \dots, \gamma_r \in \Lambda_{n,\mathbb{R}}$, their simultaneous Hodge locus is defined by the holomorphic equations

$$\gamma_a \cdot \omega_j(b) = 0, \qquad 1 \le a \le r, \quad 0 \le j \le n-2.$$

These are $r(n-1)$ equations, but their differentials need not be independent on an elliptic surface deformation space. Section 3 constructs suitable integral classes and nearby surfaces at which all these differentials are independent.

3. Geometric realization and density of Mordell–Weil rank jumps

We construct transverse intersections of the local period image of $\mathcal{W}_n^{\mathrm{sb}}$ with simultaneous fixed-class Hodge loci in $\mathcal{D}_n^+$. We then identify the resulting integral Hodge classes with independent Mordell–Weil sections. Fix a height $n \ge 2$, and put

$$h := n-1, \qquad N := 10n-2.$$

The all-I_1 locus $\mathcal{W}_n^{\mathrm{I}_1}$ is obtained from a non-empty Zariski-open subset of the coefficient space

$$H^0(\mathbb{P}^1, \mathcal{O}(4n)) \oplus H^0(\mathbb{P}^1, \mathcal{O}(6n))$$

by quotienting by Weierstrass scaling and changes-of-coordinates on the base [PS25, Thms. 1.1 and 1.2(a)]. The coefficient space is irreducible, so its non-empty Zariski-open subset and the quotient $\mathcal{W}_n^{\mathrm{I}_1}$ are irreducible. Moreover,

$$\dim \mathcal{W}_n^{\mathrm{I}_1} = (4n+1) + (6n+1) - 1 - 3 = 10n-2.$$

We use the primitive lattice and period equations of Section 2. The construction applies to every positive integer r satisfying $rh \le N$; neither the integral classes nor their Gram matrices are prescribed in advance.

3.1. **The simply-branched locus.**

We first identify a dense open locus on which each ramification point supplies one deformation parameter. Let

$$\varphi_S \colon \mathbb{P}^1 \longrightarrow \overline{\mathcal{M}}_{1,1} \simeq \mathcal{P}(4,6)$$

be the classifying morphism of $[S] \in \mathcal{W}_n^{\mathrm{I}_1}$ to the weighted projective stack $\mathcal{P}(4,6)$. Following [SB23, §2], we write $\mathrm{Ram}(\varphi_S)$ for the zero divisor of the stack-theoretic differential

$$\mathrm{d}\varphi_S \colon T_{\mathbb{P}^1} \longrightarrow \varphi_S^* T_{\overline{\mathcal{M}}_{1,1}}.$$

This differs from the ramification divisor of the coarse j-map at the orbifold values $j = 0, 1728$. The all-I_1 condition makes φ_S non-constant and unramified over the boundary.

By [SB23, Lem. 2.1 and Prop. 2.8(4)],

$$T_{\overline{\mathcal{M}}_{1,1}} \simeq \mathcal{O}(10), \qquad \varphi_S^* \mathcal{O}(1) \simeq \mathcal{O}_{\mathbb{P}^1}(n).$$

Thus $\mathrm{d}\varphi_S$ is a non-zero section of $\mathcal{O}_{\mathbb{P}^1}(10n-2)$, and

$$\deg \mathrm{Ram}(\varphi_S) = 10n - 2 = N.$$

For Weierstrass coefficients $a_i \in H^0(\mathbb{P}^1, \mathcal{O}(in))$, $i = 4, 6$, this divisor is cut out, including multiplicities, by the weighted Wronskian

$$\Theta(a_4, a_6) \coloneqq 3a_6\, \mathrm{d}a_4 - 2a_4\, \mathrm{d}a_6 \in H^0\big(\mathbb{P}^1, \mathcal{O}(10n) \otimes \Omega^1_{\mathbb{P}^1}\big).$$

Here the differentials are computed in local frames of $\mathcal{O}(n)$. Under a change of frame by a unit u, the terms involving $\mathrm{d}u$ cancel, giving $\Theta(u^{-4}a_4, u^{-6}a_6) = u^{-10}\Theta(a_4, a_6)$; hence the expression defines the stated global section. This agrees, up to a non-zero constant, with Shepherd–Barron's expression for the ramification divisor [SB23, §10, Lem. 10.2]. Since a_4 and a_6 have no common zero, the infinitesimal scaling vector $(4a_4, 6a_6)$ is nowhere zero. Taking its determinant with $(\mathrm{d}a_4, \mathrm{d}a_6)$ represents the differential by $4a_4\, \mathrm{d}a_6 - 6a_6\, \mathrm{d}a_4 = -2\Theta(a_4, a_6)$, so the zero divisors agree, including multiplicities.

Definition 3.1. A surface $[S] \in \mathcal{W}_n^{\mathrm{I}_1}$ is *simply-branched* if the stack-theoretic ramification divisor $\mathrm{Ram}(\varphi_S)$ is reduced. We denote the corresponding substack by

$$\mathcal{W}_n^{\mathrm{sb}} \subset \mathcal{W}_n^{\mathrm{I}_1}.$$

Thus, after passing to an étale neighbourhood which orders the ramification divisor, a point of $\mathcal{W}_n^{\mathrm{sb}}$ has

$$\mathrm{Ram}(\varphi_S) = p_1 + \cdots + p_N$$

with the p_i pairwise distinct. This definition does not require the images $\varphi_S(p_i)$ to be pairwise distinct.

Lemma 3.2. *The substack $\mathcal{W}_n^{\mathrm{sb}}$ is Zariski open and dense in $\mathcal{W}_n^{\mathrm{I}_1}$.*

Proof. Reducedness of the zero divisor of $\Theta(a_4, a_6)$ is a Zariski-open condition on the Weierstrass coefficients. Thus $\mathcal{W}_n^{\mathrm{sb}}$ is Zariski open in $\mathcal{W}_n^{\mathrm{I}_1}$. To prove density, we show that multiple ramification points can be split by deforming the surface.

We use Shepherd–Barron's local deformation theory. The height n, base-genus-zero part of his stack $\mathcal{JE}^{\mathrm{gen}}$ is our all-I_1 locus [SB23, p. 2523]. Its classifying morphisms are non-constant and unramified over the boundary (see the local calculation in the proof of [HP19, Prop. 11]); their ramification divisors need not be reduced. His numerical hypotheses hold because the base has genus zero and $p_g = n - 1 > 0$. Moreover, the relatively-minimal surface coincides with its minimal Weierstrass model, since there are no fibre components disjoint from the zero section.

Here $\mathrm{Def}(S)$ denotes the miniversal analytic deformation space of the Jacobian elliptic fibration with its zero section, with the base considered up to isomorphism. It underlies a local quotient chart of $\mathcal{W}_n^{\mathrm{I}_1}$ at $[S]$ [SB23, Prop. 2.9(2)–(3) and Prop. 4.4(1)].

Suppose that the ramification divisor of a given surface has multiplicity $e_i - 1$ at a point p_i. Shepherd–Barron's local isomorphism [SB23, Prop. 4.4] identifies this

miniversal chart with the product of the deformation spaces of the finite schemes

$$\operatorname{Spec}\mathbb{C}[t]/(t^{e_i}).$$

The i-th local deformation base has dimension $e_i - 1$. Since

$$\sum_i (e_i - 1) = \deg \operatorname{Ram}(\varphi_S) = N,$$

their product is an N-dimensional miniversal chart. On the simply-branched locus, each factor is one-dimensional, giving one deformation parameter for each ramification point. More generally, the product description allows us to deform the ramification at the different points independently. To split a point of multiplicity $e-1$, lift φ_S locally to an étale chart of the target stack and choose coordinates w on the source and s on that chart. Shepherd–Barron's Schiffer deformation has the form [SB23, Prop. 4.3]

$$s = w^e + e\sum_{j=0}^{e-2} t_{e-j} w^j.$$

Up to the non-zero factor e, its critical divisor is therefore cut out by

$$w^{e-1} + \sum_{j=1}^{e-2} j\, t_{e-j} w^{j-1}.$$

For $e = 2$ this divisor is already reduced. For $e \geq 3$, setting $t_{e-1} \neq 0$ and all other relevant parameters equal to zero gives

$$w^{e-1} + t_{e-1},$$

which has distinct roots. Thus the discriminant of the critical polynomial, viewed as a function of the deformation parameters, is not identically zero. Choose pairwise disjoint coordinate discs around the points p_i. By Rouché's theorem, after shrinking the deformation chart each disc contains exactly $e_i - 1$ critical points, counted with multiplicity. These account for the total ramification degree N, so there are no critical points outside their union. The simultaneous non-vanishing of the local discriminants defines a dense open subset of the miniversal chart. Thus every surface in $\mathcal{W}_n^{\mathrm{I}_1}$ is an analytic limit of surfaces with reduced ramification.

Since $\mathcal{W}_n^{\mathrm{I}_1}$ is irreducible by the quotient description in [PS25, Thms. 1.1 and 1.2(a)], its non-empty Zariski-open substack $\mathcal{W}_n^{\mathrm{sb}}$ is Zariski dense. ∎

Lemma 3.3. *For every $n \geq 2$, both inclusions*

$$\mathcal{W}_n^{\mathrm{sb}} \subset \mathcal{W}_n^{\mathrm{I}_1} \subset \mathcal{W}_{\min,n}$$

are Zariski open and dense. In particular, $\mathcal{W}_n^{\mathrm{sb}}$ is analytically dense in $\mathcal{W}_{\min,n}$.

Proof. Consider the irreducible coefficient space

$$V_n = H^0(\mathbb{P}^1, \mathcal{O}(4n)) \oplus H^0(\mathbb{P}^1, \mathcal{O}(6n)).$$

Minimality is a Zariski-open condition on the Weierstrass data [BPS24, Prop. 4.15]. Intersecting with the open condition $\Delta \not\equiv 0$ gives a non-empty Zariski-open subset of V_n presenting $\mathcal{W}_{\min,n}$ after quotienting by Weierstrass scaling and changes-of-coordinates on the base [PS25, Thms. 1.1–1.2].

Reducedness of the discriminant divisor on $\mathbb{P}^1$ is also a Zariski-open condition. This condition is non-empty: in homogeneous coordinates $[s : t]$ on $\mathbb{P}^1$, take

$$a_4 = -3s^{4n}, \qquad a_6 = t^{6n}.$$

Then

$$\Delta = -16(4a_4^3 + 27a_6^2) = -432(t^{12n} - 4s^{12n})$$

has $12n$ distinct zeros, and it does not vanish at $[s : t] = [0 : 1]$. These Weierstrass data are minimal. Indeed, non-minimality at a point would force the discriminant to vanish there to order at least 12.

Thus the squarefree-discriminant locus is a non-empty Zariski-open subset of the irreducible minimal elliptic coefficient locus. Passing to the quotient shows that $\mathcal{W}_n^{\mathrm{I}_1}$ is Zariski open and dense in $\mathcal{W}_{\min,n}$.

Lemma 3.2 gives the same conclusion for $\mathcal{W}_n^{\mathrm{sb}} \subset \mathcal{W}_n^{\mathrm{I}_1}$. Hence $\mathcal{W}_n^{\mathrm{sb}}$ is Zariski open and dense in $\mathcal{W}_{\min,n}$. A non-empty Zariski-open subset of an irreducible complex variety is analytically dense; applying this on the coefficient presentation gives the analytic assertion. ∎

3.2. **The Shepherd–Barron frame.** Fix $[S] \in \mathcal{W}_n^{\mathrm{sb}}$ and order its ramification points as $p_1, \dots, p_N$. Choose an affine coordinate z on the base for which none of the p_i lies at infinity, and put $z_i = z(p_i)$. On the corresponding affine Weierstrass chart, write

$$y^2 = x^3 + a_4(z)x + a_6(z), \qquad \omega_0 = \frac{\mathrm{d}x \wedge \mathrm{d}z}{2y}.$$

The canonical bundle formula identifies global holomorphic two-forms with polynomials in z of degree at most $n-2$, multiplied by ω_0. Thus

$$H^{2,0}(S) = \bigoplus_{j=0}^{h-1} \mathbb{C}\,\omega_j, \qquad \omega_j = z^j \omega_0.$$

The form ω_0 is nowhere vanishing over the affine z-line. We use this polynomial identification for sections of $\mathcal{O}_{\mathbb{P}^1}(n-2)$ below; in particular, $f(p_i)$ means the scalar $f(z_i)$.

Every automorphism of the Jacobian fibration preserves its zero section and the reduced discriminant divisor of degree $12n$. Its induced automorphism of $\mathbb{P}^1$ therefore belongs to the finite group preserving these $12n \geq 3$ points, while the kernel embeds in the finite group $\mathrm{Aut}_{\mathbb{C}(t)}(E_S, 0)$. Thus

$$G := \mathrm{Aut}_{\mathcal{W}_n^{\mathrm{sb}}}([S])$$

is finite. Since $\mathcal{W}_n^{\mathrm{sb}}$ is an open substack of $\mathcal{W}_{\min,n}$, it is smooth and separated Deligne–Mumford by [PS25, Thm. 1.2(a)–(b)]. By [ES18, Prop. 2.9], it has an analytic open neighbourhood of the form $[Z/K]$ with K finite. Restricting near a lift of $[S]$ to its stabilizer G and linearizing the finite group action gives an analytic open immersion

$$[B/G] \longrightarrow \mathcal{W}_n^{\mathrm{sb}},$$

where B is a simply-connected analytic ball and $0 \in B$ lifts $[S]$ and is fixed by G. The quotient B/G identifies with an analytic neighbourhood of the image of $[S]$ in the coarse moduli space.

The relative ramification divisor is finite étale of degree N over B. Since B is simply-connected, the ordering at 0 extends to an ordering throughout B. We trivialize the primitive integral local system over B by flat transport. By the preceding deformation-space identification and [SB23, §4, Prop. 4.4], after shrinking B there are analytic coordinates

$$(t_1, \dots, t_N)$$

such that $\partial/\partial t_i|_0$ spans the Schiffer direction associated with p_i. Shepherd–Barron associates to each p_i a meromorphic two-form $\widetilde{\eta}_i$ with poles of order at most two along the fibre over p_i and no other poles. It is of the second kind, meaning that its residue along that fibre vanishes. Its class is taken in the primitive quotient $F^\perp/\mathbb{C}F$. The zero section identifies this quotient with $\Lambda_{n,\mathbb{C}}$ by

$$[x] \longmapsto x - (x \cdot O)F.$$

This expression is independent of the representative $x \in F^\perp$ and is orthogonal to both F and O. Taking its $(1,1)$-component gives

$$\eta_i = \mathrm{pr}^{1,1}[\widetilde{\eta}_i] \in H^{1,1}_{\mathrm{prim}}(S).$$

Shepherd–Barron's meromorphic classes $[\widetilde{\eta}_i]$ lie in $H^{2,0}(S) \oplus H^{1,1}_{\mathrm{prim}}(S)$ [SB23, Thm. 4.10]. The $H^{2,0}(S)$-summand pairs trivially with this space, so

$$[\widetilde{\eta}_i] \cdot [\widetilde{\eta}_j] = \eta_i \cdot \eta_j.$$

Since the ramification divisor is reduced, the projected classes $\eta_1, \dots, \eta_N$ therefore form an orthogonal basis of $H^{1,1}_{\mathrm{prim}}(S)$ for the intersection pairing [SB23, Thm. 4.11(4)].

Write $E_{p_i} = \pi^{-1}(p_i)$ for the fibre over p_i. Let F^2 denote the holomorphic Hodge bundle over B, whose fibre at b is $H^{2,0}(S_b)$. The Gauss–Manin connection ∇ differentiates sections of the cohomology bundle using its flat trivialization. For every $\omega \in H^{2,0}(S)$, Shepherd–Barron's rank-one formula [SB23, Thm. 1.4(3)] gives

$$\left(\nabla_{\partial/\partial t_i}\omega\right)^{1,1} = c_i(\omega)\eta_i.$$

The derivative is evaluated at $0 \in B$ using any local holomorphic extension of ω in F^2. Its $(1,1)$-projection is independent of that extension: changing the extension only adds an element of $H^{2,0}(S)$ to the derivative; see [CMSP17, Lem. 5.3.2]. Here c_i is a linear functional with

$$\ker(c_i) = H^0(S, \Omega^2_S(-E_{p_i})).$$

Since

$$K_S \simeq \pi^* \mathcal{O}_{\mathbb{P}^1}(n-2),$$

the canonical linear system is base-point free for $n \geq 2$. Under the polynomial identification above, vanishing along E_{p_i} is the single condition $f(p_i) = 0$. Hence this kernel is a hyperplane, and c_i is non-zero.

Griffiths transversality places the derivative in $H^{2,0}(S) \oplus H^{1,1}_{\mathrm{prim}}(S)$; Shepherd–Barron's formula further shows that, for each Schiffer direction, the $(1,1)$-components of the derivatives of all holomorphic two-forms lie in the same line $\mathbb{C}\eta_i$. The kernel description identifies c_i with evaluation at p_i, up to a non-zero scalar, giving the following formula.

Lemma 3.4. *For $f \in H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(n-2))$ one has*

$$c_i(f\,\omega_0) = c_i(\omega_0) f(p_i), \qquad c_i(\omega_0) \neq 0.$$

In particular,

$$c_i(\omega_j) = c_i(\omega_0) z_i^j, \qquad 0 \leq j \leq h-1.$$

Proof. Under the polynomial identification above, write $f = f(p_i) + (f - f(p_i))$. Since $f - f(p_i)$ vanishes at p_i, the form $(f - f(p_i))\omega_0$ belongs to $\ker(c_i)$. Linearity therefore gives

$$c_i(f\,\omega_0) = f(p_i) c_i(\omega_0).$$

Since ω_0 does not vanish along E_{p_i}, it does not belong to $\ker(c_i)$, so $c_i(\omega_0) \neq 0$. Taking $f(z) = z^j$ gives the final formula. ∎

The evaluation formula lets us test whether the h period equations for a fixed class have independent differentials. The following criterion does this by selecting h ramification points.

Theorem 3.5. *Let γ be a locally flat real class which is of type $(1,1)$ at $[S] \in \mathcal{W}_n^{\mathrm{sb}}$. If there are distinct indices $i_1, \ldots, i_h$ such that*

$$\gamma \cdot \eta_{i_k} \neq 0 \qquad (1 \leq k \leq h),$$

then, on the marked local deformation space, the fixed-class type-$(1,1)$ locus defined by γ is smooth of codimension $h = n-1$ at $[S]$.

Proof. Extend $\omega_0, \ldots, \omega_{h-1}$ to a holomorphic frame of F^2 near $[S]$ and set

$$f_j = \gamma \cdot \omega_j.$$

For a real flat class, vanishing of these pairings is equivalent to being of type $(1,1)$; the conjugate equations follow automatically. Flatness of γ and compatibility of the Gauss–Manin connection with the intersection pairing give $\partial f_j / \partial t_i = \gamma \cdot \nabla_{\partial/\partial t_i} \omega_j$. At $[S]$, the $H^{2,0}(S)$-component of the derivative pairs trivially with γ, since γ has type $(1,1)$. Shepherd–Barron's formula and Lemma 3.4 therefore give

$$\left.\frac{\partial f_j}{\partial t_{i_k}}\right|_{[S]} = c_{i_k}(\omega_0) z_{i_k}^j (\gamma \cdot \eta_{i_k}).$$

If J_γ denotes this $h \times h$ minor, then

$$\det J_\gamma = \left(\prod_{k=1}^{h} c_{i_k}(\omega_0)(\gamma \cdot \eta_{i_k}) \right) \prod_{1 \leq a < b \leq h} (z_{i_b} - z_{i_a}).$$

The first product is non-zero by Lemma 3.4 and the hypothesis on γ. The second is non-zero because the ramification points are distinct. Hence the Jacobian of $(f_0, \ldots, f_{h-1})$ has rank h at $[S]$. These equations define the fixed-class type-$(1,1)$ locus, so the holomorphic implicit function theorem makes this locus smooth of codimension h at $[S]$. ∎

The same period calculation shows that every non-zero fixed integral class in Λ_n has a proper Hodge locus on a marked local chart. Together with the algebraicity of Hodge loci, this gives the following well-known statement. For $n \geq 3$, it follows from Cox's result that a very general surface has Picard number two [Cox90,

Prop. 3.2]; for $n = 2$, it is classical. We include a proof using Schiffer deformation coordinates.

Proposition 3.6. *For every $n \geq 2$, a very general member of $\mathcal{W}_{\min,n}$ has Mordell–Weil rank zero.*

Proof. At any point of $\mathcal{W}_n^{\mathrm{sb}}$, every non-zero integral Hodge class in Λ_n pairs non-trivially with some Schiffer image class. The rank-one formula therefore gives a deformation direction in which one of its periods has non-zero derivative. Consequently, every non-zero fixed-class Hodge locus on a marked local chart is a proper closed analytic subset. Since the lattice is countable, Baire's theorem shows that their union has empty analytic interior.

Apply the algebraicity theorem of Cattani–Deligne–Kaplan [CDK95, Thm. 1.1 and Cor. 1.3] to the primitive variation on a smooth algebraic atlas of finite type for $\mathcal{W}_n^{\mathrm{sb}}$. The locus carrying a non-zero integral Hodge class is a countable union of closed algebraic subsets of this atlas. Their images in $\mathcal{W}_n^{\mathrm{sb}}$ are constructible. None is dense, since a dense constructible image would contain a non-empty open subset, contradicting the local empty-interior assertion.

Their closures in $\mathcal{W}_{\min,n}$, together with the proper closed complement of $\mathcal{W}_n^{\mathrm{sb}}$, form a countable union of proper closed substacks. Outside this union there are no non-zero integral Hodge classes in Λ_n. Since all singular fibres are irreducible, the Shioda–Tate formula gives Mordell–Weil rank zero. ∎

For the height bounds, we need a stronger local avoidance statement: on a relatively compact marked chart, only finitely many non-zero integral classes with bounded $-\gamma^2/2$ can become Hodge.

Lemma 3.7. *Let B_0 be a marked local deformation chart in $\mathcal{W}_n^{\mathrm{sb}}$, and let $B \subset B_0$ be a connected open subset whose closure is compact and contained in B_0. For every $H > 0$, the locus*

$$B_{\leq H} = \left\{ b \in B : \text{there exists } 0 \neq \gamma \in \Lambda_n \cap H^{1,1}(S_b) \text{ with } -\frac{1}{2}\gamma^2 \leq H \right\}$$

is a proper closed analytic subset of B. Its complement $B_{>H}$ is therefore open and dense.

Proof. Fix a Euclidean norm on $\Lambda_{n,\mathbb{R}}$. The primitive real $(1,1)$-spaces vary continuously over $\overline{B}$, and the intersection pairing is negative definite on each of them. Compactness therefore gives a constant $c > 0$ such that

$$-\gamma^2 \geq c\|\gamma\|^2 \qquad \left(b \in \overline{B},\ \gamma \in H^{1,1}_{\mathrm{prim}}(S_b, \mathbb{R})\right).$$

Consequently, every non-zero integral Hodge class with $-\gamma^2/2 \leq H$ belongs to the finite set

$$A_H = \left\{ \gamma \in \Lambda_n : 0 < -\frac{1}{2}\gamma^2 \leq H,\ \|\gamma\|^2 \leq \frac{2H}{c} \right\}.$$

Thus $B_{\leq H}$ is the union of the fixed-class Hodge loci of the classes in A_H. Each is closed analytic and proper by the period calculation in the proof of Proposition 3.6. Their finite union is proper because B is connected. ∎

3.3. **Simultaneous period equations.** For r classes, we need rh period equations with independent differentials. We first construct a complex tuple for which the Jacobian splits into r Vandermonde blocks, then use polynomial non-vanishing to obtain a real tuple.

Lemma 3.8. *Let $r \geq 1$, put $M = rh$, and suppose $M \leq N$. There exist $\mathbb{R}$-linearly independent vectors*

$$v_1, \dots, v_r \in W := H^{1,1}_{\mathrm{prim}}(S, \mathbb{R}), \qquad v_a^2 = -1,$$

and a subset $I \subset \{1, \dots, N\}$ of cardinality M such that the $M \times M$ matrix

$$J_{(a,j),i} = c_i(\omega_0) z_i^j (v_a \cdot \eta_i), \qquad 1 \leq a \leq r, \quad 0 \leq j \leq h-1, \quad i \in I,$$

is non-singular.

Proof. Choose any M indices and partition them into disjoint blocks

$$I = C_1 \sqcup \cdots \sqcup C_r, \qquad |C_a| = h.$$

Over $\mathbb{C}$, set

$$v_a^\circ = \sum_{i \in C_a} \frac{\eta_i}{\eta_i^2}.$$

The denominators are non-zero because the η_i form an orthogonal basis of a non-degenerate quadratic space. Orthogonality gives

$$v_a^\circ \cdot \eta_i = \begin{cases} 1, & i \in C_a, \\ 0, & i \notin C_a. \end{cases}$$

After ordering rows and columns by the blocks, the resulting matrix is block diagonal. Its a-th block has determinant

$$\left(\prod_{i \in C_a} c_i(\omega_0) \right) \prod_{\substack{i,k \in C_a \\ i<k}} (z_k - z_i),$$

which is non-zero.

Thus the determinant is a non-zero polynomial on

$$\left(H^{1,1}_{\mathrm{prim}}(S) \right)^r = (W \otimes_{\mathbb{R}} \mathbb{C})^r.$$

A complex polynomial which vanishes on the real form W^r vanishes identically. Hence we may choose a real tuple on which the determinant is non-zero. The chosen vectors are linearly independent: any relation $\sum_a \lambda_a v_a = 0$ would give the same relation among the rows indexed by $(a, 0)$, contradicting determinant non-vanishing. Finally, negative definiteness allows us to replace each v_a by $v_a / \sqrt{-v_a^2}$. This multiplies its h corresponding rows by the same non-zero scalar, so the determinant remains non-zero. ∎

To pass from the real tuple to integral classes, we approximate its directions by indivisible lattice vectors of increasing norm. These approximations may leave the original Hodge subspace; we will subsequently deform the surface to make the chosen integral classes Hodge.

Lemma 3.9. *Let $L \subset E$ be a full lattice, meaning a discrete subgroup spanning the finite-dimensional real vector space E. Assume $\dim E \geq 2$ and equip E with a Euclidean norm. Let $v_1, \dots, v_r \in E$ be linearly independent unit vectors. There exist indivisible vectors*

$$\gamma_{a,m} \in L, \qquad 1 \leq a \leq r,$$

such that

$$\|\gamma_{a,m}\| \longrightarrow \infty, \qquad \frac{\gamma_{a,m}}{\|\gamma_{a,m}\|} \longrightarrow v_a,$$

and, for all sufficiently large m, the vectors $\gamma_{1,m}, \dots, \gamma_{r,m}$ are linearly independent. If a proper real subspace $E_0 \subsetneq E$ contains all the v_a, the vectors $\gamma_{a,m}$ may moreover be chosen outside E_0.

Proof. After choosing a basis of L, the directions of indivisible lattice vectors are precisely the rational directions, which are dense in the unit sphere of E. For each m, only finitely many indivisible vectors have norm at most m; deleting their finitely many directions does not destroy density. If $E_0 \subsetneq E$ is a proper real subspace, rational directions outside E_0 are still dense: every open subset of the unit sphere contains a smaller open subset disjoint from E_0, and rational directions are dense there. For each a and m, we may therefore choose an indivisible vector $\gamma_{a,m}$, outside E_0 when specified, such that

$$\|\gamma_{a,m}\| > m, \qquad \left\| \frac{\gamma_{a,m}}{\|\gamma_{a,m}\|} - v_a \right\| < \frac{1}{m}.$$

These inequalities give the two asserted limits. Since linear independence is an open condition on tuples of vectors, convergence to the independent tuple $(v_1, \dots, v_r)$ gives independence for all sufficiently large m. ∎

3.4. **Local simultaneous realization.** The implicit function theorem lets us compensate for a small change in the cohomology classes by a small deformation of the surface. We apply this to indivisible lattice vectors whose normalized directions approach the real tuple constructed above, while their norms tend to infinity.

Theorem 3.10. *Let $n \geq 2$, let $[S] \in \mathcal{W}_n^{\mathrm{sb}}$, and let $r \geq 1$ satisfy*

$$r(n-1) \leq 10n - 2.$$

Then there are points $b_m \to 0$ in the marked local deformation chart B chosen above, representing pairwise non-isomorphic Jacobian elliptic surfaces,

$$[S_m] \in \mathcal{W}_n^{\mathrm{sb}}, \qquad [S_m] \longrightarrow [S],$$

and, after identifying the primitive cohomology lattices by flat transport, indivisible classes

$$\gamma_{1,m}, \dots, \gamma_{r,m} \in \Lambda_n$$

with the following properties:

(1) *every $\gamma_{a,m}$ is of type $(1,1)$ on S_m;*
(2) *the classes $\gamma_{1,m}, \dots, \gamma_{r,m}$ span a negative-definite subspace and are linearly independent;*

(3) *on the marked local deformation space, the* $r(n-1)$ *period equations defining their simultaneous fixed-class Hodge locus have linearly independent differentials at* $[S_m]$*; in particular, this locus is smooth of complex codimension* $r(n-1)$ *there;*
(4) *for every* a*,*

$$d_{a,m} := -\frac{\gamma_{a,m}^2}{2} \in \mathbb{Z}_{>0}, \qquad d_{a,m} \longrightarrow \infty.$$

Proof. Put $M = rh$. Work on the simply-connected chart B above. Flat transport identifies the real primitive cohomology of its fibres with $\Lambda_{n,\mathbb{R}}$. At the central surface, Hodge decomposition gives an orthogonal splitting

$$\Lambda_{n,\mathbb{R}} = H_+ \oplus W,$$

where

$$H_+ = \big(H^{2,0}(S) \oplus H^{0,2}(S)\big) \cap \Lambda_{n,\mathbb{R}}, \qquad W = H^{1,1}_{\mathrm{prim}}(S,\mathbb{R}).$$

The intersection form is positive definite on H_+ and negative definite on W. We therefore use the positive-definite norm determined by

$$\|v+w\|^2 = v^2 - w^2, \qquad v \in H_+, \quad w \in W.$$

In particular, the vectors supplied by Lemma 3.8 have norm one.

Choose $v_1,\dots,v_r \in W$ and $I \subset \{1,\dots,N\}$ as in Lemma 3.8. Let $s = (t_i)_{i\in I}$ be the corresponding M coordinates, and let $B_I \subset B$ be the slice on which all other coordinates are zero. Restrict B_I to a sufficiently small coordinate polydisc $\mathbb{D}^M \subset B$ with coordinates s, where $\mathbb{D} \subset \mathbb{C}$ is an open disc centred at zero. Choose a holomorphic frame

$$\omega_0(t),\dots,\omega_{h-1}(t)$$

of F^2 over B whose value at the origin is the monomial basis above. With the notation $\omega_j(s) = \omega_j(t)|_{B_I}$, define the restriction of the Hodge-equation map to this slice

$$\Psi\colon (\Lambda_{n,\mathbb{C}})^r \times \mathbb{D}^M \longrightarrow \mathbb{C}^{rh}$$

by

$$\Psi_{a,j}(\boldsymbol{u},s) = u_a \cdot \omega_j(s), \qquad 1 \le a \le r, \quad 0 \le j \le h-1.$$

Because $v_a \in W$, one has $\Psi(\boldsymbol{v},0) = 0$. Moreover, the same calculation as in the proof of Theorem 3.5, using Lemma 3.4, gives

$$\left.\frac{\partial \Psi_{a,j}}{\partial t_i}\right|_{(\boldsymbol{v},0)} = c_i(\omega_0) z_i^j (v_a \cdot \eta_i),$$

so the derivative with respect to s is non-singular by Lemma 3.8.

The holomorphic implicit function theorem therefore gives a neighbourhood $\mathcal{V}$ of $\boldsymbol{v} = (v_1,\dots,v_r)$ in $(\Lambda_{n,\mathbb{C}})^r$ and a holomorphic map

$$\tau\colon \mathcal{V} \longrightarrow \mathbb{D}^M, \qquad \tau(\boldsymbol{v}) = 0,$$

such that

$$\Psi(\boldsymbol{u},\tau(\boldsymbol{u})) = 0 \tag{3}$$

for every $\boldsymbol{u} \in \mathcal{V}$. Shrinking $\mathcal{V}$ if necessary, the derivative with respect to s remains non-singular along the graph of τ.

The period equations are homogeneous in each class, so normalization does not change their zero locus. Since $h > 0$, the positive subspace H_+ is non-zero and $W \subset \Lambda_{n,\mathbb{R}}$ is proper. We may therefore apply Lemma 3.9 to the lattice $\Lambda_n \subset \Lambda_{n,\mathbb{R}}$ with $E_0 = W$. Choose indivisible vectors $\gamma_{a,m} \notin W$ such that

$$u_{a,m} := \frac{\gamma_{a,m}}{\|\gamma_{a,m}\|} \longrightarrow \nu_a, \qquad \|\gamma_{a,m}\| \longrightarrow \infty.$$

For large m, the tuple $\boldsymbol{u}_m$ lies in $\mathcal{V}$. Let $b_m \in B$ have selected coordinates $\tau(\boldsymbol{u}_m)$ and all remaining coordinates zero, and let S_m be the corresponding surface. Then $b_m \to 0$, and (3) yields

$$\gamma_{a,m} \cdot \omega = 0 \qquad \text{for every } \omega \in F^2(S_m) \quad \text{and every } 1 \le a \le r.$$

The classes $\gamma_{a,m}$ are real, so they are also orthogonal to $\overline{F^2(S_m)}$ and hence are of type $(1,1)$.

For the fixed tuple $(\gamma_{1,m}, \dots, \gamma_{r,m})$, the simultaneous Hodge locus in the full chart B is defined by the rh equations

$$\gamma_{a,m} \cdot \omega_j(t) = 0, \qquad 1 \le a \le r, \quad 0 \le j \le h-1.$$

Their Jacobian contains the selected $M \times M$ minor from the implicit function argument. Replacing $u_{a,m}$ by $\gamma_{a,m}$ multiplies each of the h rows belonging to a by $\|\gamma_{a,m}\|$, so this minor remains non-singular. The full Jacobian therefore has rank $M = rh$, and the simultaneous fixed-class Hodge locus is smooth of codimension M at $[S_m]$.

Each $\gamma_{a,m}$ is a non-zero class in $H^{1,1}_{\mathrm{prim}}(S_m, \mathbb{R})$, where the intersection pairing is negative definite. Thus their span is negative definite, and each $d_{a,m}$ is positive. By Lemma 3.9, the classes are linearly independent for all sufficiently large m. Each $d_{a,m}$ is an integer because Λ_n is even. Since $v_a^2 = -1$, one has

$$\frac{d_{a,m}}{\|\gamma_{a,m}\|^2} = -\frac{\gamma_{a,m}^2}{2\|\gamma_{a,m}\|^2} \longrightarrow \frac{1}{2}.$$

Therefore $d_{a,m} \to \infty$.

Finally, $\gamma_{a,m} \notin W$ implies that it is not of type $(1,1)$ at the central surface, so $\tau(\boldsymbol{u}_m) \neq 0$. These non-zero points tend to zero; after passing to a subsequence they are pairwise distinct. Since G is finite, a further subsequence lies in pairwise distinct G-orbits. By the choice of the uniformizing chart, their images in the coarse analytic moduli space are pairwise distinct; equivalently, the corresponding Jacobian elliptic surfaces are pairwise non-isomorphic. All the resulting points lie in $\mathcal{W}_n^{\mathrm{sb}}$ by construction. After discarding finitely many initial terms and relabelling the final subsequence, properties (1)–(4) hold for every m. ∎

3.5. **Mordell–Weil rank, heights, and analytic density.** Since every fibre is irreducible, the Shioda–Tate description removes only the fibre and zero-section classes. The primitive integral Hodge classes constructed above therefore give Mordell–Weil sections, and their intersection pairing determines the canonical heights. By avoiding bounded-height Hodge classes before applying local realization, we make the least non-zero section height tend to infinity. We then use density of the simply-branched locus to obtain statements throughout the minimal elliptic locus.

Theorem 3.11. *Let n, r be integers satisfying $n \geq 2$ and*

$$1 \leq r \leq \left\lfloor \frac{10n-2}{n-1} \right\rfloor.$$

For every $[S] \in \mathcal{W}_n^{\mathrm{sb}}$ there are pairwise non-isomorphic Jacobian elliptic surfaces

$$[S_m] \in \mathcal{W}_n^{\mathrm{sb}}, \qquad [S_m] \longrightarrow [S],$$

and linearly independent sections

$$P_{1,m}, \ldots, P_{r,m} \in \mathrm{MW}(S_m/\mathbb{P}^1)$$

such that

$$\min_{0 \neq P \in \mathrm{MW}(S_m/\mathbb{P}^1)} \widehat{h}(P) \longrightarrow \infty.$$

In particular,

$$\mathrm{rk}\,\mathrm{MW}(S_m/\mathbb{P}^1) \geq r.$$

The corresponding Shioda classes are indivisible in Λ_n, and the canonical heights of non-zero sections are positive integers. On the marked local deformation space, the $r(n-1)$ period equations defining the simultaneous fixed-class Hodge locus of the chosen classes have linearly independent differentials at $[S_m]$; hence this locus is smooth of complex codimension $r(n-1)$ there.

Proof. Choose a marked local deformation chart B centred at a point b_0 representing $[S]$, relatively compact in a larger marked chart, and a decreasing neighbourhood basis U_m of b_0. By Lemma 3.7, each $U_m \cap B_{>m}$ is non-empty and open. Choose a point in this intersection and apply Theorem 3.10 on a sufficiently small marked deformation chart contained in $U_m \cap B_{>m}$. Its sequence contains infinitely many pairwise non-isomorphic surfaces, and all sufficiently late terms have representatives in $U_m \cap B_{>m}$. Choose one such representative b_m, with $S_m = S_{b_m}$ not isomorphic to any previously chosen surface, together with its r independent indivisible integral Hodge classes $\gamma_{1,m}, \ldots, \gamma_{r,m}$. Use the marking on B to regard these classes as elements of Λ_n. Then $b_m \to b_0$, and the simultaneous period differentials are independent at b_m. Put

$$d_{a,m} = -\frac{1}{2}\gamma_{a,m}^2.$$

Since $b_m \in B_{>m}$, we have $d_{a,m} > m$ for every a.

Since S_m has singular fibres, its Néron–Severi group is torsion-free [Shi90, Thm. 1.2]. We therefore identify $\mathrm{NS}(S_m)$ with its image in $H^2(S_m, \mathbb{Z})/\mathrm{tors}$. The Lefschetz $(1,1)$-theorem gives $\gamma_{a,m} \in \mathrm{NS}(S_m)$ for every a [CMSP17, §3.4, pp. 119–120].

All fibres of S_m are irreducible, so $\mathrm{Triv}(S_m) = \langle F, O \rangle$. Since this lattice is unimodular and F and O are algebraic, the integral orthogonal splitting from Section 2 restricts to

$$\mathrm{NS}(S_m) = \mathrm{Triv}(S_m) \oplus \big(\mathrm{NS}(S_m) \cap \Lambda_n\big).$$

The Shioda–Tate description [Shi90, Thm. 1.3] therefore gives

$$\mathrm{MW}(S_m/\mathbb{P}^1) \simeq \mathrm{NS}(S_m)/\,\mathrm{Triv}(S_m) \simeq \mathrm{NS}(S_m) \cap \Lambda_n.$$

In particular, $\mathrm{MW}(S_m/\mathbb{P}^1)$ is torsion-free. Under this identification, the map from sections to algebraic classes in Λ_n is Shioda's orthogonal projection σ_{Sh} [Shi90, Lem. 8.1]. Consequently, each $\gamma_{a,m}$ determines a unique section $P_{a,m}$ with

$$\sigma_{\mathrm{Sh}}(P_{a,m}) = \gamma_{a,m}.$$

Since this correspondence is an isomorphism of abelian groups, the linear independence of the classes gives the linear independence of the sections.

Shioda's height pairing [Shi90, Thm. 8.4], with our half-normalization, gives

$$\widehat{h}(P_{a,m}) = -\frac{1}{2}\,\sigma_{\mathrm{Sh}}(P_{a,m})^2 = -\frac{1}{2}\,\gamma_{a,m}^2 = d_{a,m} \in \mathbb{Z}_{>0}, \qquad d_{a,m} \longrightarrow \infty.$$

More generally, every non-zero section $P \in \mathrm{MW}(S_m/\mathbb{P}^1)$ has a non-zero integral Shioda class of type $(1,1)$. Since $b_m \in B_{>m}$, it follows that

$$\widehat{h}(P) = -\frac{1}{2}\,\sigma_{\mathrm{Sh}}(P)^2 > m.$$

The lattice Λ_n is even and its real primitive $(1,1)$-subspace is negative definite, so these heights are positive integers. The Mordell–Weil group has positive rank, hence the set of non-zero section heights is non-empty and has a minimum. Therefore

$$\min_{0 \neq P \in \mathrm{MW}(S_m/\mathbb{P}^1)} \widehat{h}(P) > m,$$

which proves the asserted divergence.

The chosen classes were selected from applications of Theorem 3.10. Their indivisibility and the independence of their simultaneous period differentials are therefore preserved. The holomorphic implicit function theorem gives the stated smoothness and codimension of their simultaneous fixed-class Hodge locus. ∎

Corollary 3.12. *Let $n \geq 2$. Every analytic neighbourhood of every point of $\mathcal{W}_{\min,n}$ contains a sequence of pairwise non-isomorphic Jacobian elliptic surfaces $[S_m] \in \mathcal{W}_n^{\mathrm{sb}}$ with sections $P_m \in \mathrm{MW}(S_m/\mathbb{P}^1)$ whose canonical heights are strictly increasing positive integers. In particular, infinitely many positive integers occur as section heights within every such neighbourhood.*

Proof. Let U be an analytic neighbourhood of any point of $\mathcal{W}_{\min,n}$. By Lemma 3.3, choose $[S] \in U \cap \mathcal{W}_n^{\mathrm{sb}}$. Apply Theorem 3.11 at $[S]$ with $r = 1$. The resulting pairwise non-isomorphic surfaces converge to $[S]$, so all sufficiently late members lie in U. Their section heights are positive integers tending to infinity. Passing to a subsequence makes these heights strictly increasing. ∎

Corollary 3.13. *Let $n \geq 2$ and $r \geq 1$ satisfy $r(n-1) \leq 10n-2$. The locus*

$$\left\{[S] \in \mathcal{W}_n^{\mathrm{sb}} : \mathrm{rk}\,\mathrm{MW}(S/\mathbb{P}^1) \geq r\right\}$$

is analytically dense in $\mathcal{W}_{\min,n}$.

More precisely, for every non-empty analytic open subset $U \subset \mathcal{W}_{\min,n}$ and every $H > 0$, there are infinitely many pairwise non-isomorphic surfaces $[S'] \in U \cap \mathcal{W}_n^{\mathrm{sb}}$ carrying r independent sections with indivisible Shioda classes and satisfying

$$\widehat{h}(P) > H \qquad \text{for every } 0 \neq P \in \mathrm{MW}(S'/\mathbb{P}^1).$$

The $r(n-1)$ simultaneous fixed-class period equations are transverse at $[S']$. On a marked local deformation chart, their Hodge germ has a smooth representative Z of dimension $10n-2-r(n-1)$, with image in $U \cap \mathcal{W}_n^{\mathrm{sb}}$, such that

$$\widehat{h}(P) > H \qquad \text{for every } b \in Z \text{ and every } 0 \neq P \in \mathrm{MW}(S_b/\mathbb{P}^1).$$

Proof. By Lemma 3.3, choose $[S] \in U \cap \mathcal{W}_n^{\mathrm{sb}}$. Theorem 3.11 gives pairwise non-isomorphic surfaces $[S_m] \to [S]$ carrying the required independent sections and satisfying

$$\min_{0 \neq P \in \mathrm{MW}(S_m/\mathbb{P}^1)} \widehat{h}(P) \longrightarrow \infty.$$

All sufficiently late terms lie in U and satisfy the prescribed bound. Indivisibility and transversality follow from the same theorem.

At each selected surface, the integral Shioda identification shows that there is no non-zero integral primitive-cohomology Hodge class with $-\gamma^2/2 \leq H$. Lemma 3.7 therefore gives an open neighbourhood with the same property. Shrinking the simultaneous Hodge germ into this neighbourhood and into the inverse image of U gives the required representative. ∎

For $r = 1$, the density and arbitrarily large lower bounds for all non-zero section heights recover Cox's theorem [Cox90, Thm. 4.4]; on the all-I_1 locus, his height $P \cdot O$ and our normalization satisfy $\widehat{h}(P) = n + P \cdot O$. For general r, the rank-density assertion can also be obtained from Khelifa–Urbanik's criterion [KU25, Thm. 1.9(i)], whose monodromy hypothesis is verified in Lemma 3.24. We do not develop that alternative argument here. The proof above supplies explicit transverse period minors and smooth Hodge locus representatives on which all non-zero section heights satisfy a prescribed uniform lower bound. Subsection 3.6 identifies the corresponding section-moduli germs.

3.6. **Two-pointed moduli and deformations of sections.** In this subsection, we interpret the realization theorem in terms of moduli of elliptic surfaces with chosen sections. We first express the canonical height of a section as the degree of a line bundle pulled back from $\overline{\mathcal{M}}_{1,2}$. We then compare deformations of the sections with the corresponding period conditions. Inchiostro identifies $\mathcal{X} = \overline{\mathcal{M}}_{1,2}$ with the $(4,6)$-weighted blowup of $\mathcal{P}(2,3,4)$ at $[1:1:0]$ [Inc22, Thms. 1.1 and 2.10]. Put $\mathcal{Y} = \overline{\mathcal{M}}_{1,1} \simeq \mathcal{P}(4,6)$, and write

$$\phi \colon \mathcal{X} \longrightarrow \mathcal{Y}, \qquad p \colon \mathcal{X} \longrightarrow \mathcal{P}(2,3,4)$$

for the universal stable curve and the weighted blowdown, respectively. All weighted projective spaces here are stacks. The exceptional divisor D parametrizes curves whose two markings lie on a rational tail; under the universal-curve identification, it is the zero section. We write

$$\lambda = \mathcal{O}_{\mathcal{Y}}(1), \qquad \mathcal{L}_{1,2} = \phi^*\lambda, \qquad \xi = p^*\mathcal{O}_{\mathcal{P}(2,3,4)}(1).$$

Throughout this subsection, we work on the all-I_1 locus.

Proposition 3.14. *The line bundles on $\mathcal{X}$ satisfy*

$$\xi \simeq \mathcal{L}_{1,2} \otimes \mathcal{O}_{\mathcal{X}}(D), \qquad \omega_{\mathcal{X}} \simeq \mathcal{L}_{1,2}^{\otimes(-9)}. \tag{4}$$

Let $S \to \mathbb{P}^1$ represent a point of $\mathcal{W}_n^{\mathrm{I}_1}$, with classifying map φ_S and fundamental line bundle $\mathcal{L}$. A section $P \neq O$ determines a lift $f_P\colon \mathbb{P}^1 \to \mathcal{X}$ of φ_S. If $C_P = P^*O$ is the divisor where P meets O, then

$$f_P^*\mathcal{L}_{1,2} \simeq \mathcal{L}, \qquad f_P^*D = C_P, \qquad (p \circ f_P)^*\mathcal{O}(1) \simeq \mathcal{L}(C_P). \tag{5}$$

In particular,

$$\deg f_P^*\mathcal{L}_{1,2} = n, \qquad \widehat{h}(P) = \deg f_P^*\xi = n + (P \cdot O) \geq n. \tag{6}$$

Proof. We first compute the line bundles in (4). In weighted coordinates $[x : y : A]$ on $\mathcal{P}(2,3,4)$, the rational map to $\mathcal{Y}$ is given by A and $y^2 - x^3 - Ax$, of weights 4 and 6. Their pullbacks along p have exceptional factors of orders 4 and 6. Dividing by these factors gives sections of

$$\xi^{\otimes 4} \otimes \mathcal{O}_{\mathcal{X}}(-4D) = \big(\xi \otimes \mathcal{O}_{\mathcal{X}}(-D)\big)^{\otimes 4}, \qquad \xi^{\otimes 6} \otimes \mathcal{O}_{\mathcal{X}}(-6D) = \big(\xi \otimes \mathcal{O}_{\mathcal{X}}(-D)\big)^{\otimes 6}.$$

These are the two sections defining ϕ. Thus $\mathcal{L}_{1,2} \simeq \xi \otimes \mathcal{O}_{\mathcal{X}}(-D)$ in the integral Picard group of $\mathcal{X}$.

The canonical bundle of $\mathcal{P}(2,3,4)$ is $\mathcal{O}(-9)$, and the $(4,6)$-weighted blowup has discrepancy $4 + 6 - 1 = 9$. Hence

$$\omega_{\mathcal{X}} \simeq \xi^{\otimes(-9)} \otimes \mathcal{O}_{\mathcal{X}}(9D) \simeq \mathcal{L}_{1,2}^{\otimes(-9)},$$

which proves the second identity.

We now construct f_P. Since all singular fibres have type I_1, the surface S is the pullback of the universal stable one-pointed curve along φ_S. The section P therefore determines a lift to $\mathcal{X}$. A section cannot pass through a fibre node: in the local equation $xy = t$, the conditions $x(0) = y(0) = 0$ would imply $\mathrm{ord}_0(xy) \geq 2$.

At a point where P meets O, the two-pointed stabilization inserts a rational tail carrying the two markings. Since D is the universal zero section, its pullback is P^*O as an effective Cartier divisor, with the collision multiplicities included. Together with $\varphi_S^*\lambda \simeq \mathcal{L}$ and the first identity in (4), this proves (5). The map f_P lands in $\mathcal{M}_{1,2}$ precisely where the fibre is smooth and the two sections are distinct.

It remains to identify the degree with the canonical height. Put $d = n + (P \cdot O)$. The class

$$\gamma_P = [P] - [O] - d[F]$$

is orthogonal to F because $(P - O) \cdot F = 0$ and $F^2 = 0$. The coefficient d is forced by orthogonality to O, since $(P - O) \cdot O = n + P \cdot O = d$ and $F \cdot O = 1$. Since $P^2 = O^2 = -n$ and $P \cdot F = O \cdot F = 1$, we have $\gamma_P^2 = -2d$. All fibres are irreducible, so γ_P is Shioda's projection of P. Our normalization gives $\widehat{h}(P) = -\gamma_P^2/2 = d$, while (5) gives $\deg f_P^*\xi = d$. Finally, $P \neq O$ implies $P \cdot O \geq 0$, proving (6). ∎

We can also describe the map to $\mathcal{P}(2,3,4)$ directly from the Weierstrass equation. Write $P = (x_P, y_P)$ in $y^2 = x^3 + a_4x + a_6$, let s be the canonical section of $\mathcal{O}(C_P)$, and put $\mathcal{L}_P = \mathcal{L}(C_P)$. Then

$$X = s^2x_P \in H^0(\mathcal{L}_P^{\otimes 2}), \qquad Y = s^3y_P \in H^0(\mathcal{L}_P^{\otimes 3}), \qquad A' = s^4a_4 \in H^0(\mathcal{L}_P^{\otimes 4})$$

satisfy $Y^2 - X^3 - A'X = s^6a_6$ and define $p \circ f_P = [X : Y : A']$. To check that these sections have no common zero, consider first a collision of multiplicity k. The coordinates x_P and y_P have poles of orders $2k$ and $3k$, so X and Y are non-zero there and the image is $[1 : 1 : 0]$. Away from the collisions, a common zero

of X, Y, A' would imply $a_4 = a_6 = 0$, contrary to the all-I_1 hypothesis. Thus the associated line bundle is $\mathcal{L}_P = \mathcal{L}(C_P)$.

We next introduce the moduli stack of surfaces with several chosen sections. For an ordered tuple $\mathbf{d} = (d_1, \dots, d_r)$ of positive integers, let $\mathcal{S}^{\mathrm{I}_1}_{n,\mathbf{d}}$ denote the stack of height n all-I_1 Jacobian elliptic surfaces with an ordered tuple of non-zero sections $P_1, \dots, P_r$ satisfying $\widehat{h}(P_a) = d_a$. The base is considered up to isomorphism. The sections determine a map to the fibre product

$$\mathcal{X}^{(r)} = \underbrace{\mathcal{X} \times_{\mathcal{Y}} \cdots \times_{\mathcal{Y}} \mathcal{X}}_{r \text{ factors}}.$$

Conversely, a map $f = (f_1, \dots, f_r) \colon \mathbb{P}^1 \to \mathcal{X}^{(r)}$ has a common projection $\varphi \colon \mathbb{P}^1 \to \mathcal{Y}$. Pulling back the universal curve along φ gives the elliptic fibration, and the lifts f_a give its sections. We require $\deg \varphi^* \lambda = n$, $\deg f_a^* \xi = d_a$, and the pullback of the nodal boundary to be reduced. The positivity of d_a excludes the zero section, since $\xi|_D \simeq \mathcal{O}_D$. These constructions commute with base change. Thus $\mathcal{S}^{\mathrm{I}_1}_{n,\mathbf{d}}$ is the corresponding open locus of maps of the specified degrees, modulo PGL_2. The fibre product allows the sections P_a to meet.

Let $\rho \colon \mathcal{S}^{\mathrm{I}_1}_{n,\mathbf{d}} \to \mathcal{W}^{\mathrm{I}_1}_n$ be the morphism forgetting the additional sections. Its source and target have finite stabilizers; automorphisms are required to preserve the zero section. The degrees $d_a = \deg f_a^* \xi$ are locally constant in families, so fixing $\mathbf{d}$ does not change the local deformation germ of a tuple.

Proposition 3.15. *Let $(S, P_1, \dots, P_r)$ be a point of $\mathcal{S}^{\mathrm{I}_1}_{n,\mathbf{d}}$, and let $B \to \mathcal{W}^{\mathrm{I}_1}_n$ be a simply-connected analytic étale local deformation chart through $[S]$. Choose a marking and a lift of the tuple over the central point b_0. Put $\gamma_a = [P_a] - [O] - d_a[F]$, and define*

$$\Phi(b) = \big(\gamma_a \cdot \omega_j(b)\big)_{a,j}, \qquad 1 \le a \le r, \quad 0 \le j \le n-2,$$

where $\omega_0(b), \dots, \omega_{n-2}(b)$ is a local holomorphic frame of $H^{2,0}(S_b)$. If $d\Phi$ has rank $r(n-1)$ at b_0, then $\mathcal{S}^{\mathrm{I}_1}_{n,\mathbf{d}}$ is smooth at the tuple, of dimension

$$10n - 2 - r(n-1). \tag{7}$$

After base change to B, the morphism ρ identifies the germ at the chosen lift with $(\Phi^{-1}(0), b_0)$.

Proof. We first compute the deformations of the corresponding map $f = (f_{P_1}, \dots, f_{P_r}) \colon \mathbb{P}^1 \to \mathcal{X}^{(r)}$. Each component lies in the smooth locus of ϕ. This includes collisions with O, which correspond to smooth points of the original one-pointed fibre under the universal-curve identification. Thus $\mathcal{X}^{(r)}$ is smooth in a neighbourhood of $f(\mathbb{P}^1)$. Since $P_a \simeq \mathbb{P}^1$, $K_S = (n-2)F$, and $F \cdot P_a = 1$, adjunction gives

$$P_a^2 = -2 - K_S \cdot P_a = -n.$$

Thus $N_{P_a/S} \simeq \mathcal{O}_S(P_a)|_{P_a} \simeq \mathcal{O}_{\mathbb{P}^1}(-n)$. Along a section, the vertical tangent line of $S \to \mathbb{P}^1$ identifies with the section's normal bundle. Since S is pulled back from the universal curve, base change therefore identifies $f_{P_a}^* T_{\mathcal{X}/\mathcal{Y}}$ with $N_{P_a/S}$. Together with

$$\varphi_S^* T_{\mathcal{Y}} \simeq \mathcal{L}^{\otimes 10} \simeq \mathcal{O}_{\mathbb{P}^1}(10n),$$

the relative tangent sequence gives, with $E = f^*T_{\mathcal{X}^{(r)}}$,

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^1}(-n)^{\oplus r} \longrightarrow E \longrightarrow \mathcal{O}_{\mathbb{P}^1}(10n) \longrightarrow 0. \tag{8}$$

Taking cohomology gives

$$0 \longrightarrow H^0(E) \longrightarrow H^0(\mathcal{O}_{\mathbb{P}^1}(10n)) \xrightarrow{\delta} \bigoplus_{a=1}^{r} H^1(N_{P_a/S}) \longrightarrow H^1(E) \longrightarrow 0. \tag{9}$$

For a first-order deformation of φ_S, local lifts of P_a exist because ϕ is smooth along f_{P_a}. Their differences on overlaps form a Čech cocycle with values in $N_{P_a/S}$. Its class is δ_a; it vanishes precisely when the local lifts glue to a deformation of the section.

To pass to deformations over an unparameterized base, we quotient by PGL_2. This gives

$$T_{b_0}B \simeq H^0(\varphi_S^* T_{\mathcal{Y}}) \big/ d\varphi_S\big(H^0(T_{\mathbb{P}^1})\big).$$

The denominator has dimension three, since an infinitesimal source stabilizer would preserve the $12n$ distinct discriminant points and hence vanish. These reparametrizations lift through df, so δ descends to $T_{b_0}B$.

We next compare this obstruction map with the period derivative. For a section $P = P_a$, consider

$$0 \longrightarrow \mathcal{O}_S \longrightarrow \mathcal{O}_S(P) \longrightarrow N_{P/S} \longrightarrow 0.$$

Since $H^0(N_{P/S}) = 0$, we have $h^0(\mathcal{O}_S(P)) = 1$. Also $H^2(\mathcal{O}_S(P)) = 0$ by Serre duality: the divisor $K_S - P$ has intersection -1 with the nef fibre class, so it cannot be effective. Riemann–Roch gives

$$\chi(\mathcal{O}_S(P)) = n + \tfrac{1}{2}P \cdot (P - K_S) = n + \tfrac{1}{2}(-n - (n-2)) = 1.$$

It follows that $H^1(\mathcal{O}_S(P)) = 0$. The connecting map, also called the divisor semiregularity map, is therefore an isomorphism

$$s_P \colon H^1(N_{P/S}) \xrightarrow{\sim} H^2(\mathcal{O}_S). \tag{10}$$

For the Cartier divisor $P \subset S$, the semiregularity map of Buchweitz–Flenner agrees with s_P: both are Serre dual to the restriction map $H^0(K_S) \to H^0(K_S|_P)$. The Kodaira–Spencer map associates to each tangent vector in $T_{b_0}B$ the corresponding first-order deformation class in $H^1(S, T_S)$. Compatibility with this map identifies the image of the section obstruction under s_P with the infinitesimal Hodge obstruction for $[P]$, taking values in $H^2(\mathcal{O}_S) \simeq H^{0,2}(S)$; see [BF00, Thms. 1.3, 4.5 and 5.5(1)–(2), §5.3]. Since $[O]$ and $[F]$ remain Hodge, this is also the obstruction for γ_a. Thus the composite

$$T_{b_0}B \xrightarrow{\delta} \bigoplus_{a=1}^{r} H^1(N_{P_a/S}) \xrightarrow{\oplus_a s_{P_a}} H^2(\mathcal{O}_S)^{\oplus r}$$

agrees with $d\Phi$ after applying Serre duality $H^2(\mathcal{O}_S) \simeq H^0(K_S)^\vee$ and evaluating in the chosen frame, up to the sign convention for the connecting maps. Since each s_{P_a} is an isomorphism, a first-order deformation lifts all the chosen sections if and only if their classes remain of type $(1,1)$ to first order.

The rank hypothesis now implies that δ is surjective. By (9), $H^1(E) = 0$. For maps from the fixed curve $\mathbb{P}^1$ to a smooth Deligne–Mumford stack, the tangent space is $H^0(E)$ and an obstruction space is $H^1(E)$. Indeed, local lifts across a square-zero extension differ by sections of E, and their obstruction to gluing is

the resulting Čech class in $H^1(E)$. Thus the mapping stack is smooth at f. Using (8), its dimension is

$$h^0(E) = \chi(E) = (10n+1) + r(1-n).$$

Quotienting by PGL_2 subtracts 3, proving (7). In particular, for $r = 1$ under the rank hypothesis, the actual dimension equals the expected dimension $9n-1$, independently of d_1.

It remains to identify the germs. The morphism ρ is representable: for a fixed surface family, the tuples of sections form an open locus in a product of relative Hilbert spaces. Hence the germ H of $\mathcal{S}^{\mathrm{I}_1}_{n,\mathbf{d}} \times_{\mathcal{W}^{\mathrm{I}_1}_n} B$ at the chosen lift is an analytic space, smooth by the preceding argument.

The divisors P_a preserve their marked integral classes in this family. Thus the period functions Φ vanish on H, and, since H is reduced, ρ factors holomorphically through $\Phi^{-1}(0)$. Let h_0 denote the chosen lift in H. The obstruction sequence and the quotient by source reparametrizations give

$$T_{h_0}H \simeq H^0(E)\big/ df\big(H^0(T_{\mathbb{P}^1})\big) \xrightarrow{d\rho} \ker\big(d\Phi(b_0)\big) \subset T_{b_0}B,$$

where the displayed arrow is an isomorphism. There are no additional tangent directions from moving the sections on a fixed surface because $H^0(N_{P_a/S}) = 0$. Both germs are smooth of dimension $10n-2-r(n-1)$, so the analytic inverse function theorem identifies them. Smoothness here is asserted for $\mathcal{S}^{\mathrm{I}_1}_{n,\mathbf{d}}$ and its local charts. ∎

The sections constructed in Theorem 3.11 satisfy the transversality hypothesis of the preceding proposition. We therefore obtain the following moduli-theoretic interpretation of the realization theorem.

Corollary 3.16. *Let $n \geq 2$ and $r \geq 1$ be integers satisfying $r(n-1) \leq 10n-2$. For every $[S] \in \mathcal{W}^{\mathrm{sb}}_n$, the surfaces and sections with indivisible Shioda classes of Theorem 3.11 give points*

$$(S_m, P_{1,m}, \dots, P_{r,m}) \in \mathcal{S}^{\mathrm{I}_1}_{n,\mathbf{d}_m}, \qquad d_{a,m} = \widehat{h}(P_{a,m}) \longrightarrow \infty.$$

At each such point, the stack is smooth of dimension $10n-2-r(n-1)$, and its germ on the chosen marked chart is the simultaneous fixed-class Hodge locus germ. Moreover,

$$\deg f^*_{P_{a,m}} \mathcal{L}_{1,2} = n, \qquad \deg(p \circ f_{P_{a,m}})^* \mathcal{O}(1) = d_{a,m}, \qquad (P_{a,m} \cdot O_m) = d_{a,m} - n.$$

Every analytic neighbourhood of every point of $\mathcal{W}_{\min,n}$ contains infinitely many pairwise non-isomorphic surfaces with these smooth section-moduli germs. For any prescribed $H > 0$, the surfaces and representatives of their germs may be chosen so that every non-zero Mordell–Weil section on every parametrized surface has canonical height greater than H.

Proof. Theorem 3.11 supplies independent sections with indivisible Shioda classes with heights tending to infinity and full-rank period derivatives. Proposition 3.15 gives the smoothness and the identification of germs, while Proposition 3.14 gives the degree formulas.

The final assertion follows from Corollary 3.13, using Proposition 3.15 to identify the resulting fixed-class Hodge germs with the section-moduli germs. ∎

We now combine the marked local construction, the height bounds, and the section-moduli comparison to complete the proofs of the main realization theorems.

Proof of Theorems 1.2 and 1.3. The construction in the proof of Theorem 3.11 takes place in a single marked local deformation chart B centred at b_0 representing $[S]$. It gives points $b_m \to b_0$ representing pairwise non-isomorphic surfaces, independent sections with indivisible Shioda classes, divergence of the least non-zero section height, and independence of the simultaneous period differentials.

Proposition 3.14 identifies the Shioda classes with $[P_{a,m}]-[O_m]-d_{a,m}[F_m]$, where $d_{a,m} = \widehat{h}(P_{a,m})$. Proposition 3.15 therefore gives the asserted smoothness, dimension, and identification of germs. Fixing the heights does not change these germs, since the heights are locally constant in families with chosen sections; restricting to the open substack $\mathcal{W}_n^{\mathrm{sb}}$ likewise does not change them.

Corollary 3.13 gives the assertions in arbitrary analytic neighbourhoods, including the uniform lower bounds for all non-zero section heights on representatives of the Hodge germs. Taking $r = 10$ gives dimension $10n-2-10(n-1) = 8$. After shrinking each representative, the germ comparison extends the 10 chosen sections over it. Their Shioda classes remain flat, so their height-pairing matrix is constant. ∎

Remark 3.17. The number of additional sections has a natural interpretation in the birational geometry of pointed genus one curves. The open locus in $\mathcal{X}^{(r)}$ parametrizing smooth elliptic curves with $O, P_1, \dots, P_r$ pairwise distinct identifies with $\mathcal{M}_{1,r+1}$. Thus $\mathcal{X}^{(r)}$ is birational to $\overline{\mathcal{M}}_{1,r+1}$: the zero section contributes one marking. Writing κ for Kodaira dimension, the compactified coarse moduli spaces satisfy

$$\kappa\big(\overline{M}_{1,r+1}\big) = \begin{cases} -\infty, & 1 \le r \le 9, \\ 0, & r = 10, \\ 1, & r \ge 11, \end{cases}$$

by [BF06, §4, Thm. 3 and the preceding discussion].

The same threshold appears in the canonical bundle and expected deformation dimension of our fiber product target. Let $q\colon \mathcal{X}^{(r)} \to \mathcal{Y}$ be the common projection, and let $U \subset \mathcal{X}^{(r)}$ be the open locus where each coordinate lies in the relative smooth locus of ϕ. Since $\omega_{\mathcal{Y}} \simeq \lambda^{\otimes(-10)}$ and $\omega_{\mathcal{X}} \simeq \phi^*\lambda^{\otimes(-9)}$, we have

$$\omega_{\mathcal{X}/\mathcal{Y}} \simeq \phi^*\lambda, \qquad \omega_U \simeq (q|_U)^*\lambda^{\otimes(r-10)}.$$

The locus U contains the images of the section maps considered above, including their collisions. For a height n map $f : \mathbb{P}^1 \to U$, the bundle $E = f^*T_U$ therefore has rank $r+1$ and degree $(10-r)n$. Riemann–Roch gives the expected dimension, after quotienting by source automorphisms,

$$\chi(E) - 3 = (10-r)n + r - 2 = 10n - 2 - r(n-1).$$

Under the transversality hypothesis of Proposition 3.15, this is the actual local dimension. In particular, at $r = 10$ the canonical bundle of U is trivial and the dimension is $8 = \dim U - 3$, independently of n. At $r = 11$ the expected dimension

is $9-n$, so positive-dimensional families at height $n \geq 9$ necessarily exceed the expected dimension.

The canonical-bundle identity concerns the fibre-product stack on U; the stable pointed compactification has additional boundary terms.

3.7. **Exact Mordell–Weil ranks.** The realization argument can be sharpened to give exact Mordell–Weil ranks outside countable unions of proper closed analytic subsets of suitable Hodge loci. We first give an explicit criterion for rank one, then prove exact-rank-r density for $1 \leq r \leq 9$ when $n \geq 3$ and for $1 \leq r \leq 18$ when $n = 2$. The underlying infinitesimal argument is classical: compare conditions (1) and (3) and the proof of [Voi07, Thm. 3.1]. The density step is a local form of Green's density criterion, recalled in [Voi07, Prop. 3.3]. Shepherd–Barron's frame makes the required kernel conditions explicit. In rank one, non-vanishing against every Schiffer image class suffices; for several classes, we construct a non-empty open set of suitable tuples using Vandermonde blocks.

Keep $n \geq 2$, $h = n-1$, and $N = 10n-2$. Work on a sufficiently small full marked deformation chart B in $\mathcal{W}_n^{\mathrm{sb}}$, centred at 0, and write S_b for the surface over $b \in B$. Use the Schiffer tangent basis, the distinct finite ramification coordinates $z_1, \ldots, z_N$, and the basis $\eta_1, \ldots, \eta_N$ of $H^{1,1}_{\mathrm{prim}}(S_0)$ constructed above. Extend $\omega_0, \ldots, \omega_{h-1}$ to a holomorphic frame of F^2 over B. For a flat real class $\gamma \in \Lambda_{n,\mathbb{R}}$, define

$$\Phi_\gamma(b) = \big(\gamma \cdot \omega_0(b), \ldots, \gamma \cdot \omega_{h-1}(b)\big), \qquad Z_\gamma = \Phi_\gamma^{-1}(0). \tag{11}$$

Since γ is real, Z_γ is precisely its type-$(1,1)$ locus.

If γ is Hodge at 0, Lemma 3.4 identifies its period Jacobian in the Schiffer tangent basis as

$$\mathrm{d}\Phi_\gamma(0) = A_\gamma = V \operatorname{diag}\big(c_i(\omega_0)(\gamma \cdot \eta_i)\big)_{i=1}^{N}, \qquad V = (z_i^j)_{\substack{0 \leq j < h \\ 1 \leq i \leq N}}. \tag{12}$$

We keep the non-zero factors $c_i(\omega_0)$ from the preceding construction.

Lemma 3.18. *Let $z_1, \ldots, z_N$ be distinct complex numbers, let $h \geq 1$, and assume $N \geq h$ and $N > 2h-2$. Set $V = (z_i^j)_{0 \leq j < h,\, 1 \leq i \leq N}$. For $a \in (\mathbb{C}^*)^N$ and $b \in \mathbb{C}^N$, put*

$$A = V \operatorname{diag}(a), \qquad C = V \operatorname{diag}(b).$$

If $\ker A \subseteq \ker C$, then $b = ca$ for some $c \in \mathbb{C}$.

Proof. The first h columns of V have determinant

$$\prod_{1 \leq i < k \leq h} (z_k - z_i) \neq 0,$$

so A has row rank h. Every row of C annihilates $\ker A$, hence belongs to the row space of A. Applying this to the first row gives a polynomial p of degree at most $h-1$ such that

$$b_i = a_i p(z_i) \qquad (1 \leq i \leq N). \tag{13}$$

For $h = 1$, this proves the assertion.

For $h \geq 2$, apply the same argument to the last row of C. There is a polynomial q of degree at most $h-1$ satisfying

$$z_i^{h-1} b_i = a_i q(z_i) \qquad (1 \leq i \leq N).$$

Dividing by $a_i \neq 0$ and using (13), we find that $z^{h-1}p(z) - q(z)$ vanishes at all N points z_i. Its degree is at most $2h-2 < N$, so it vanishes identically. If $p \neq 0$, the identity $z^{h-1}p = q$ gives

$$h - 1 + \deg p = \deg q \leq h - 1.$$

Thus p is constant. If $p = 0$, then $b = 0$. In either case $b = ca$. ∎

Proposition 3.19. *Let $\alpha \in \Lambda_n$ be non-zero, indivisible in Λ_n, and of type $(1,1)$ on S_0. Assume*

$$\alpha \cdot \eta_i \neq 0 \qquad (1 \leq i \leq N). \tag{14}$$

Then Z_α is smooth of complex codimension $h = n-1$ at 0. For every $\beta \in \Lambda_n \smallsetminus \mathbb{Q}\alpha$ that is Hodge at 0, one has

$$\mathrm{d}\Phi_\beta(0)\big|_{T_0 Z_\alpha} \neq 0. \tag{15}$$

After replacing Z_α by a sufficiently small connected representative of its germ, there is a countable union $\mathcal{E}$ of proper closed analytic subsets of Z_α such that

$$\Lambda_n \cap H^{1,1}(S_b) = \mathbb{Z}\alpha, \qquad \mathrm{rk}\,\mathrm{MW}(S_b/\mathbb{P}^1) = 1 \qquad (b \in Z_\alpha \smallsetminus \mathcal{E}). \tag{16}$$

These points are dense in Z_α and have full measure for every measure absolutely continuous with respect to intrinsic smooth volume on Z_α.

Proof. Condition (14) allows us to apply Theorem 3.5 to α. Its Jacobian calculation shows that $\mathrm{d}\Phi_\alpha(0)$ is surjective, so Z_α is smooth of codimension h and

$$T_0 Z_\alpha = \ker A_\alpha. \tag{17}$$

Shrink the chart so that Z_α is a connected complex manifold.

Let $\beta \in \Lambda_n \smallsetminus \mathbb{Q}\alpha$ be Hodge at 0. If (15) failed, then

$$\ker A_\alpha \subseteq \ker A_\beta.$$

Apply Lemma 3.18 with

$$a_i = c_i(\omega_0)(\alpha \cdot \eta_i), \qquad b_i = c_i(\omega_0)(\beta \cdot \eta_i).$$

The numerical hypotheses hold since

$$N \geq h, \qquad N - (2h-2) = 8n + 2 > 0.$$

The lemma gives $b_i = ca_i$ for some $c \in \mathbb{C}$. Cancelling $c_i(\omega_0) \neq 0$, we obtain

$$(\beta - c\alpha) \cdot \eta_i = 0 \qquad (1 \leq i \leq N).$$

Both α and β lie in $H^{1,1}_{\mathrm{prim}}(S_0)$. The intersection pairing on this space is non-degenerate: its real form is negative definite by the Hodge index theorem. Since the η_i form a basis, the displayed equalities force $\beta = c\alpha$ in $\Lambda_{n,\mathbb{C}}$. Comparing an integral lattice coordinate in which α is non-zero gives $c \in \mathbb{Q}$, a contradiction. This proves (15).

For each $\beta \in \Lambda_n \smallsetminus \mathbb{Q}\alpha$, the subset $Z_\alpha \cap Z_\beta$ is closed analytic in Z_α. If β is not Hodge at 0, this subset does not contain 0 and is therefore proper. If β is Hodge at 0, equation (15) shows that Φ_β cannot vanish identically on Z_α, so the subset is again proper. Thus

$$\mathcal{E} = \bigcup_{\beta \in \Lambda_n \smallsetminus \mathbb{Q}\alpha} (Z_\alpha \cap Z_\beta)$$

is a countable union of proper closed analytic subsets. Each is nowhere dense and has measure zero with respect to every smooth positive density on Z_α. Baire's theorem gives density of the complement, and countable additivity gives the measure assertion.

For $b \notin \mathcal{E}$, every integral Hodge class in Λ_n belongs to $\mathbb{Q}\alpha$. Since α is indivisible, $\Lambda_n \cap \mathbb{Q}\alpha = \mathbb{Z}\alpha$. The integral Shioda identification established in the proof of Theorem 3.11, valid for every surface in $\mathcal{W}_n^{\mathrm{sb}}$, therefore gives

$$\mathrm{MW}(S_b/\mathbb{P}^1) \simeq \Lambda_n \cap H^{1,1}(S_b) = \mathbb{Z}\alpha.$$

This proves (16). ∎

Corollary 3.20. *For every $n \geq 2$, the locus*

$$\left\{[S] \in \mathcal{W}_n^{\mathrm{sb}} : \mathrm{rk}\,\mathrm{MW}(S/\mathbb{P}^1) = 1\right\}$$

is dense in $\mathcal{W}_{\min,n}$ for the complex analytic topology.

Proof. Let $U \subset \mathcal{W}_{\min,n}$ be a non-empty analytic open subset. By Lemma 3.3, choose $[S_0] \in U \cap \mathcal{W}_n^{\mathrm{sb}}$ and a full marked deformation chart B centred at 0 whose image lies in $U \cap \mathcal{W}_n^{\mathrm{sb}}$.

Each functional $\lambda \mapsto \lambda \cdot \eta_i$ on $H^{1,1}_{\mathrm{prim}}(S_0, \mathbb{R})$ is non-zero, by non-degeneracy of the intersection pairing and $\eta_i \neq 0$. Avoiding their finitely many proper kernels, choose

$$\lambda \in H^{1,1}_{\mathrm{prim}}(S_0, \mathbb{R}), \qquad \lambda \cdot \eta_i \neq 0 \quad (1 \leq i \leq N). \tag{18}$$

By (12), the derivative in the B variables of

$$\Psi_1 : \Lambda_{n,\mathbb{C}} \times B \longrightarrow \mathbb{C}^h, \qquad \Psi_1(v, b) = \big(v \cdot \omega_j(b)\big)_{0 \leq j < h} \tag{19}$$

is surjective at $(\lambda, 0)$. Applying the holomorphic implicit function theorem on an h-coordinate slice, as in the proof of Theorem 3.10, gives a holomorphic map τ_1 near λ with $\tau_1(\lambda) = 0$ and $\Psi_1(v, \tau_1(v)) = 0$. Choose non-zero vectors $v_m \in \Lambda_n \otimes_{\mathbb{Z}} \mathbb{Q}$ converging to λ in the domain of τ_1, and put $b_m = \tau_1(v_m)$. Then $b_m \to 0$ and v_m is Hodge on S_{b_m}.

By [SB23, Prop. 5.1], the primitive $(1,1)$-Hodge bundle is the direct image of a line bundle on the relative ramification divisor. Over the ordered chart B, this divisor splits into N copies of B, giving line subbundles whose fibres are the Schiffer image lines. Using the smooth Hodge splitting, after shrinking B we may choose continuous non-zero representatives $\eta_i(b)$ with $\eta_i(0) = \eta_i$. Here the smooth Hodge splitting is the smoothly varying decomposition $\Lambda_{n,\mathbb{C}} = H^{2,0}(S_b) \oplus H^{1,1}_{\mathrm{prim}}(S_b) \oplus H^{0,2}(S_b)$ in the marking. Hence

$$v_m \cdot \eta_i(b_m) \longrightarrow \lambda \cdot \eta_i \neq 0.$$

For sufficiently large m, an indivisible generator α_m of $\Lambda_n \cap \mathbb{Q}v_m$ therefore satisfies the hypotheses of Proposition 3.19 at b_m. Its one-class Hodge locus contains exact-rank-one points arbitrarily close to b_m, whose images lie in U. Since U was arbitrary, the assertion follows. ∎

We now extend the argument to several classes. The following lemma constructs tuples for which the only additional period conditions vanishing to first

order along the simultaneous Hodge locus come from linear combinations of the imposed classes.

Lemma 3.21. *Let $z_1,\dots,z_N$ be distinct complex numbers, and let $h,r \geq 1$ satisfy $rh \leq N$. Assume that either $h=1$ or $r(h+1)\leq N$. Put*

$$V=(z_i^j)_{\substack{0\leq j<h,\\ 1\leq i\leq N}}, \qquad B_x = V\,\mathrm{diag}(x) \qquad (x\in\mathbb{C}^N).$$

For $\boldsymbol{x}=(x^1,\dots,x^r)\in(\mathbb{C}^N)^r$, set

$$\mathcal{A}_{\boldsymbol{x}}=\begin{pmatrix} B_{x^1}\\ \vdots \\ B_{x^r}\end{pmatrix}, \qquad \mathcal{K}_{\boldsymbol{x}}=\left\{y\in\mathbb{C}^N : \ker\mathcal{A}_{\boldsymbol{x}}\subseteq\ker B_y\right\}.$$

There is a non-empty Zariski-open subset $\mathcal{U}\subset(\mathbb{C}^N)^r$ such that every $\boldsymbol{x}\in\mathcal{U}$ satisfies

$$\mathrm{rk}\,\mathcal{A}_{\boldsymbol{x}}=rh, \qquad \mathcal{K}_{\boldsymbol{x}}=\mathrm{span}_{\mathbb{C}}\{x^1,\dots,x^r\}. \tag{20}$$

Proof. If $h=1$, the rows of $\mathcal{A}_{\boldsymbol{x}}$ are $x^1,\dots,x^r$. For every linearly independent tuple, the annihilator of $\ker\mathcal{A}_{\boldsymbol{x}}$ is their row space, giving (20). Such tuples form a non-empty Zariski-open subset because $r\leq N$.

For exact rank, each block must also detect whether an additional period condition vanishes along the simultaneous Hodge locus. We therefore use blocks of size $h+1$: the extra column leaves a one-dimensional kernel, and vanishing on this kernel will force the additional coefficient vector to be proportional to the imposed one on that block. Suppose $h\geq 2$. Choose disjoint subsets

$$C_1,\dots,C_r\subset\{1,\dots,N\}, \qquad |C_a|=h+1,$$

and let x^a be the indicator vector of C_a. After ordering the columns by these blocks, the non-zero blocks of $\mathcal{A}_{\boldsymbol{x}}$ are the matrices V_{C_a}, each of rank h. Hence $\mathrm{rk}\,\mathcal{A}_{\boldsymbol{x}}=rh$.

For each a, the kernel of V_{C_a} is one-dimensional. Choose a non-zero generator ρ^a. Every coordinate of ρ^a is non-zero: otherwise it would give a non-trivial relation among at most h Vandermonde columns, which are linearly independent. Extend ρ^a by zero outside C_a; the resulting vector belongs to $\ker\mathcal{A}_{\boldsymbol{x}}$.

Let $y\in\mathcal{K}_{\boldsymbol{x}}$. If i lies outside all the blocks, the coordinate vector supported at i belongs to $\ker\mathcal{A}_{\boldsymbol{x}}$. Its image under B_y has first coordinate y_i, so $y_i=0$. On each block, the condition $B_y\rho^a=0$ gives

$$V_{C_a}\,\mathrm{diag}(y|_{C_a})\rho^a=0.$$

Since $\ker V_{C_a}=\mathbb{C}\rho^a$, there is a scalar c_a such that

$$\mathrm{diag}(y|_{C_a})\rho^a=c_a\rho^a.$$

All coordinates of ρ^a are non-zero, so $y_i=c_a$ for every $i\in C_a$. Thus $y=\sum_a c_a x^a$. Conversely, every linear combination of the x^a belongs to $\mathcal{K}_{\boldsymbol{x}}$, so this tuple satisfies (20).

It remains to show that the two conditions persist on a Zariski-open neighbourhood of this tuple. We express the restriction of B_y to $\ker\mathcal{A}_{\boldsymbol{x}}$ in coordinates, so that the required kernel dimension becomes a matrix-rank condition. Choose a set I of rh columns for which $(\mathcal{A}_{\boldsymbol{x}})_I$ is invertible at the constructed tuple, and put

$J = \{1, \dots, N\} \smallsetminus I$. On the open set where this minor remains invertible, define the linear map

$$R_{\boldsymbol{x}} : \mathbb{C}^N \longrightarrow \mathrm{Mat}_{h,N-rh}(\mathbb{C}), \qquad R_{\boldsymbol{x}}(y) = (B_y)_J - (B_y)_I (\mathcal{A}_{\boldsymbol{x}})_I^{-1} (\mathcal{A}_{\boldsymbol{x}})_J.$$

The kernel of $\mathcal{A}_{\boldsymbol{x}}$ consists of the vectors whose coordinates satisfy

$$u_I = -(\mathcal{A}_{\boldsymbol{x}})_I^{-1} (\mathcal{A}_{\boldsymbol{x}})_J u_J.$$

Consequently,

$$\mathcal{K}_{\boldsymbol{x}} = \ker R_{\boldsymbol{x}}.$$

Full row rank of $\mathcal{A}_{\boldsymbol{x}}$ implies that the x^a are linearly independent. Their span always lies in $\ker R_{\boldsymbol{x}}$, so $\mathrm{rk}\, R_{\boldsymbol{x}} \leq N - r$. At the constructed tuple, equality holds. The entries of $R_{\boldsymbol{x}}$ are rational functions of $\boldsymbol{x}$ with denominators given by powers of the chosen invertible minor. Non-vanishing of a suitable $(N-r) \times (N-r)$ minor therefore defines a non-empty Zariski-open neighbourhood on which $\mathrm{rk}\, R_{\boldsymbol{x}} = N - r$. This proves the lemma. ∎

For the remaining arguments, extend Φ_γ complex-linearly to flat classes $\gamma \in \Lambda_{n,\mathbb{C}}$, and write

$$A_{\gamma,b} = \mathrm{d}\Phi_\gamma(b) \colon T_b B \longrightarrow \mathbb{C}^h.$$

For a tuple $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_r)$, put

$$\Phi_{\boldsymbol{\alpha}} = (\Phi_{\alpha_1}, \dots, \Phi_{\alpha_r}), \qquad Z_{\boldsymbol{\alpha}} = \bigcap_{a=1}^{r} Z_{\alpha_a}, \qquad \mathcal{A}_{\boldsymbol{\alpha},b} = \mathrm{d}\Phi_{\boldsymbol{\alpha}}(b).$$

Proposition 3.22. *Let $\alpha_1, \dots, \alpha_r \in \Lambda_n$ be linearly independent classes of type $(1,1)$ on S_0, and put*

$$L_{\mathbb{Q}} = \mathrm{span}_{\mathbb{Q}}\{\alpha_1, \dots, \alpha_r\}, \qquad L = \Lambda_n \cap L_{\mathbb{Q}}.$$

Suppose that $\mathcal{A}_{\boldsymbol{\alpha},0}$ has rank rh and that

$$\left\{ \beta \in H^{1,1}_{\mathrm{prim}}(S_0) : A_{\beta,0}\big|_{\ker \mathcal{A}_{\boldsymbol{\alpha},0}} = 0 \right\} = \mathrm{span}_{\mathbb{C}}\{\alpha_1, \dots, \alpha_r\}. \tag{21}$$

Then $Z_{\boldsymbol{\alpha}}$ is smooth of complex codimension rh at 0. For every $\beta \in \Lambda_n \smallsetminus L_{\mathbb{Q}}$ that is Hodge at 0, one has

$$\mathrm{d}\Phi_\beta(0)\big|_{T_0 Z_{\boldsymbol{\alpha}}} \neq 0.$$

After replacing $Z_{\boldsymbol{\alpha}}$ by a sufficiently small connected representative of its germ, there is a countable union $\mathcal{E}$ of proper closed analytic subsets such that

$$\Lambda_n \cap H^{1,1}(S_b) = L, \qquad \mathrm{rk}\, \mathrm{MW}(S_b/\mathbb{P}^1) = r \qquad (b \in Z_{\boldsymbol{\alpha}} \smallsetminus \mathcal{E}).$$

The complement of $\mathcal{E}$ is dense and has full measure for every measure absolutely continuous with respect to smooth volume on $Z_{\boldsymbol{\alpha}}$.

Proof. The rank hypothesis and the holomorphic implicit function theorem give smoothness and

$$T_0 Z_{\boldsymbol{\alpha}} = \ker \mathcal{A}_{\boldsymbol{\alpha},0}.$$

Shrink to a connected representative. If an integral class β is Hodge at 0 and its period derivative vanishes on this tangent space, condition (21) gives

$$\beta = \sum_{a=1}^{r} c_a \alpha_a \qquad (c_a \in \mathbb{C}).$$

Choose an integral basis of Λ_n and an invertible $r \times r$ minor of the coordinate matrix with columns $\alpha_1, \dots, \alpha_r$. Solving the corresponding equations for the c_a shows that every c_a is rational. Hence $\beta \in L_{\mathbb{Q}}$, proving the first-order assertion.

For every $\beta \in \Lambda_n \smallsetminus L_{\mathbb{Q}}$, the subset $Z_{\boldsymbol{\alpha}} \cap Z_\beta$ is closed analytic and proper. If β is not Hodge at 0, it misses the centre; otherwise properness follows from the first-order assertion. Set

$$\mathcal{E} = \bigcup_{\beta \in \Lambda_n \smallsetminus L_{\mathbb{Q}}} \big(Z_{\boldsymbol{\alpha}} \cap Z_\beta\big).$$

As in the proof of Proposition 3.19, countability of the lattice, Baire's theorem, and the measure-zero property of proper analytic subsets give the density and measure assertions.

All the α_a remain Hodge on $Z_{\boldsymbol{\alpha}}$, so every element of L is Hodge there. Outside $\mathcal{E}$, no integral class outside $L_{\mathbb{Q}}$ is Hodge. Thus the primitive integral Hodge lattice is exactly L. The lattice L has rank r and is saturated in Λ_n, meaning that Λ_n / L is torsion-free. The integral Shioda identification therefore gives the asserted Mordell–Weil rank. ∎

Corollary 3.23. *Let $n \geq 2$, and let r be an integer satisfying*

$$\begin{cases} 1 \leq r \leq 18, & n = 2, \\ 1 \leq r \leq 9, & n \geq 3. \end{cases}$$

Then the locus

$$\big\{[S] \in \mathcal{W}_n^{\mathrm{sb}} : \mathrm{rk}\,\mathrm{MW}(S/\mathbb{P}^1) = r\big\}$$

is analytically dense in $\mathcal{W}_{\min,n}$.

More precisely, for every non-empty analytic open subset $U \subset \mathcal{W}_{\min,n}$ and every $H > 0$, there is a smooth connected local simultaneous fixed-class Hodge locus Z of dimension

$$10n - 2 - r(n-1),$$

defined on a marked deformation chart by r independent indivisible integral classes, with image in $U \cap \mathcal{W}_n^{\mathrm{sb}}$, such that the Mordell–Weil rank is exactly r outside a countable union of proper closed analytic subsets of Z and

$$\widehat{h}(P) > H \qquad \text{for every } b \in Z \text{ and every } 0 \neq P \in \mathrm{MW}(S_b/\mathbb{P}^1).$$

Proof. Let $U \subset \mathcal{W}_{\min,n}$ be non-empty and analytically open, and let $H > 0$. By Lemma 3.3, choose a relatively compact marked deformation chart with image in $U \cap \mathcal{W}_n^{\mathrm{sb}}$. Lemma 3.7 gives a non-empty open subset on which no non-zero integral primitive-cohomology Hodge class has $-\gamma^2/2 \leq H$. Choose a sufficiently small full marked deformation chart B inside this subset, centred at 0, and use the notation above.

Put $W = H^{1,1}_{\mathrm{prim}}(S_0, \mathbb{R})$. The map

$$\kappa : W \otimes_{\mathbb{R}} \mathbb{C} \longrightarrow \mathbb{C}^N, \qquad \kappa(v) = \big(c_i(\omega_0)(v \cdot \eta_i)\big)_{i=1}^N$$

is an isomorphism because the intersection pairing is non-degenerate, the η_i form a basis, and every $c_i(\omega_0)$ is non-zero.

If $n = 2$, then $h = 1$ and $r \leq N = 18$. If $n \geq 3$, then

$$r(h+1) = rn \leq 9n \leq 10n - 2 = N.$$

Lemma 3.21 therefore applies. Pulling its non-empty Zariski-open subset back through κ gives a non-empty Zariski-open subset of $(W \otimes_{\mathbb{R}} \mathbb{C})^r$. Since the real form W^r is Zariski dense, choose a real tuple

$$\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_r) \in W^r$$

whose image satisfies (20). The tuple is linearly independent. By (12), its simultaneous period Jacobian has rank rh, and the kernel condition translates into

$$\Big\{\beta \in H^{1,1}_{\mathrm{prim}}(S_0) : A_{\beta,0}\big|_{\ker \mathcal{A}_{\boldsymbol{\lambda},0}} = 0\Big\} = \mathrm{span}_{\mathbb{C}}\{\lambda_1, \ldots, \lambda_r\}.$$

As in the local realization argument, we approximate $\boldsymbol{\lambda}$ by rational classes in the full cohomology space and deform the surface so that these approximations become Hodge. We must also retain the kernel condition, which will exclude additional persistent Hodge classes outside the span of the imposed classes. Consider

$$\Psi_r : (\Lambda_{n,\mathbb{C}})^r \times B \longrightarrow \mathbb{C}^{rh}, \qquad \Psi_r(\boldsymbol{\nu}, b) = \big(\nu_a \cdot \omega_j(b)\big)_{\substack{1 \le a \le r \\ 0 \le j < h}}.$$

Its derivative in the B variables is surjective at $(\boldsymbol{\lambda}, 0)$. Applying the holomorphic implicit function theorem on an rh-coordinate slice, as in the proof of Theorem 3.10, gives a holomorphic map τ_r on a neighbourhood of $\boldsymbol{\lambda}$, with

$$\tau_r(\boldsymbol{\lambda}) = 0, \qquad \Psi_r(\boldsymbol{\nu}, \tau_r(\boldsymbol{\nu})) = 0.$$

Choose tuples

$$\boldsymbol{\nu}_m = (\nu_{1,m}, \ldots, \nu_{r,m}) \in (\Lambda_n \otimes_{\mathbb{Z}} \mathbb{Q})^r, \qquad \boldsymbol{\nu}_m \longrightarrow \boldsymbol{\lambda},$$

in this neighbourhood, and put $b_m = \tau_r(\boldsymbol{\nu}_m)$. Then $b_m \to 0$, and every $\nu_{a,m}$ is Hodge on S_{b_m} because it is real and its period pairings vanish. For sufficiently large m, the tuple remains linearly independent and $\mathcal{A}_{\boldsymbol{\nu}_m, b_m}$ has rank rh.

We check that the kernel condition also persists. For tuples $\boldsymbol{\nu}$ of Hodge classes at b with $\mathrm{rk}\, \mathcal{A}_{\boldsymbol{\nu},b} = rh$, consider the complex-linear map

$$T_{\boldsymbol{\nu},b} : H^{1,1}_{\mathrm{prim}}(S_b) \longrightarrow \mathrm{Hom}_{\mathbb{C}}\big(\ker \mathcal{A}_{\boldsymbol{\nu},b}, \mathbb{C}^h\big), \qquad T_{\boldsymbol{\nu},b}(\beta) = A_{\beta,b}\big|_{\ker \mathcal{A}_{\boldsymbol{\nu},b}}.$$

The kernel of this map consists of the primitive $(1,1)$-classes whose period equations vanish to first order along the simultaneous Hodge locus. It always contains the imposed classes; we must show that no additional kernel directions appear under a small deformation. At $(\boldsymbol{\lambda}, 0)$, the map $T_{\boldsymbol{\lambda},0}$ has rank $N - r$. The smooth Hodge splitting gives local continuous frames for its domain spaces, while an invertible rh-column minor of $\mathcal{A}_{\boldsymbol{\nu},b}$ gives a continuously varying basis of $\ker \mathcal{A}_{\boldsymbol{\nu},b}$. Thus the matrix entries of $T_{\boldsymbol{\nu},b}$ vary continuously. A minor witnessing rank $N - r$ at $(\boldsymbol{\lambda}, 0)$ remains non-zero at $(\boldsymbol{\nu}_m, b_m)$ for sufficiently large m, giving $\mathrm{rk}\, T_{\boldsymbol{\nu}_m, b_m} \ge N - r$. On the other hand, the r independent classes $\nu_{a,m}$ lie in its kernel, giving $\mathrm{rk}\, T_{\boldsymbol{\nu}_m, b_m} \le N - r$. Consequently,

$$\ker T_{\boldsymbol{\nu}_m, b_m} = \mathrm{span}_{\mathbb{C}}\{\nu_{1,m}, \ldots, \nu_{r,m}\}.$$

For each a, let $\alpha_{a,m}$ be an indivisible generator of $\Lambda_n \cap \mathbb{Q}\nu_{a,m}$. Rescaling the individual classes preserves the simultaneous Hodge locus, the rank of its period Jacobian, and the kernel condition. Proposition 3.22, applied at b_m, therefore gives a smooth connected local simultaneous Hodge locus Z of dimension $N - rh$ on which the Mordell–Weil rank is exactly r outside a countable union of proper closed analytic subsets.

Shrink Z inside B. By the choice of B and the integral Shioda identification, every non-zero section on every surface parametrized by Z has canonical height greater than H. The complement of the exceptional analytic subsets is non-empty and maps into U, proving exact-rank-r density as well as the stronger assertion. ■

For $n \geq 3$, the range $r \leq 9$ in Corollary 3.23 is the dimensional limit of the complex first-order criterion (21). Indeed, the restriction map $T_{v,b}$ in the preceding proof must have rank $N-r$, while its target has dimension $h(N-rh)$. Thus, for $h > 1$,

$$N - r \leq h(N - rh) \quad \Longleftrightarrow \quad r(h+1) \leq N.$$

Since $h = n-1$ and $N = 10n-2$, this is equivalent to $r \leq 9$. Lemma 3.21 attains this range.

For this comparison, we use the Mumford–Tate terminology introduced in Subsection 3.8. A Hodge subdatum consists of a Mumford–Tate subgroup of the generic group, defined over $\mathbb{Q}$, together with a corresponding connected period domain. It is instructive to compare this infinitesimal limit with Khelifa–Urbanik's density criterion [KU25, Thm. 1.9(ii)]. Consider the Hodge subdatum whose generic rational primitive $(1,1)$-space is a fixed negative-definite r-plane. Its domain has codimension rh. On a marked local deformation chart, the Schiffer formula and Lemma 3.4 show that the period differential is injective, since the η_i form a basis and every $c_i(\omega_0)$ is non-zero. Hence the period image has dimension N, and their strong admissibility condition is $rh < N$. At $r = 10$, this holds for every $n \geq 3$, since $N - 10h = 8 > 0$. Lemma 3.24 verifies the required monodromy hypotheses. Their proof allows the constructed components to have the prescribed generic Mumford–Tate group [KU25, §1.3 and §4.4]. At a very general point of such a component, the rational primitive $(1,1)$-space is therefore exactly the chosen 10-plane. The integral Shioda identification gives Mordell–Weil rank exactly 10. These very general points are dense in each component, yielding exact-rank-10 analytic density without the complex first-order kernel condition above.

The failure of the complex first-order criterion at rank 10 can also be seen directly. At a transverse 10-class point, the imposed Hodge locus has tangent dimension 8, so the restriction map $T_{v,b}$ has source dimension $N = 10h + 8$ and target dimension $8h$. Consequently,

$$\dim_{\mathbb{C}} \ker T_{v,b} \geq N - 8h = 2h + 8 > 10 \qquad (h \geq 2).$$

Thus its kernel contains complex primitive $(1,1)$-classes outside the span of the 10 imposed classes. These additional classes need not be rational, and vanishing of their period differentials along the tangent space does not imply that they remain Hodge along the actual germ. Their existence therefore defeats the direct infinitesimal criterion without forcing additional Mordell–Weil sections.

The next subsection gives a stronger conclusion for $n \geq 9$: combining the realization theorem with Baldi–Klingler–Ullmo's algebraicity theorem produces smooth 8-dimensional 10-class germs on which the Mordell–Weil rank is exactly 10 outside an at most countable subset of each constructed Hodge locus, with simultaneous uniform lower bounds for the canonical heights of all non-zero sections.

3.8. **Monodromy and higher-rank loci.** We use Lönne's monodromy computation to apply Baldi–Klingler–Ullmo's algebraicity theorem. This will control higher-rank points on the 8-dimensional 10-class germs constructed above. We first establish the required monodromy statement.

Put

$$V_n = H^0(\mathbb{P}^1, \mathcal{O}(4n)) \oplus H^0(\mathbb{P}^1, \mathcal{O}(6n)).$$

Let $V_n^{\mathrm{sm}} \subset V_n$ be the open subset of Weierstrass data whose Weierstrass surface is smooth, and let $T = V_n^{\mathrm{sb}} \subset V_n^{\mathrm{sm}}$ be the non-empty open subset lying over $\mathcal{W}_n^{\mathrm{sb}}$ under the coefficient presentation of [PS25, Thms. 1.1 and 1.2(a)]. These spaces are smooth, irreducible, and quasi-projective. The primitive cohomology of the universal surface defines a polarizable $\mathbb{Z}$-variation of Hodge structure $\mathbb{V}$ with fibre Λ_n.

A Hodge tensor is a rational tensor of type $(0,0)$ in a tensor construction from a rational Hodge structure and its dual. The Mumford–Tate group is the common stabilizer of all such tensors. Its generic value in a variation is the group at a very general fibre. For a polarizable Hodge structure, this group is connected and reductive.

For a connected reductive algebraic group G, write G^{der} for its derived subgroup, generated algebraically by commutators, and $G^{\mathrm{ad}} = G/Z(G)$ for its adjoint quotient, where $Z(G)$ is the centre. We write $\mathrm{SO}(\Lambda_{n,\mathbb{Q}})$ for the determinant-one isometries of the intersection pairing and $\mathrm{PSO}(\Lambda_{n,\mathbb{Q}})$ for its quotient by its centre, which here is $\{\pm I\}$. A connected semisimple group is $\mathbb{Q}$-simple if it has no proper non-trivial connected normal algebraic subgroup defined over $\mathbb{Q}$.

Lemma 3.24. *Let H be the connected algebraic monodromy group of $\mathbb{V}$ over T, namely the identity component of the Zariski closure of its monodromy group, and let G be its generic Mumford–Tate group. Then*

$$H = G^{\mathrm{der}} = \mathrm{SO}(\Lambda_{n,\mathbb{Q}}).$$

In particular,

$$G^{\mathrm{ad}} \simeq \mathrm{PSO}(\Lambda_{n,\mathbb{Q}})$$

is $\mathbb{Q}$-simple.

Proof. Lönne constructs a family of smooth Weierstrass surfaces of height n, realized as double covers of the Hirzebruch surface $\mathbb{F}_{2n}$ branched along the section at infinity and a smooth trisection. The family is parametrized by the complement of a discriminant in a ball. Its monodromy is generated by the reflections in all (-2)-vectors of an even unimodular lattice of rank $12n-4$, and this lattice together with these vectors is a complete vanishing lattice [Lön02, Prop. 1]. The vanishing cycles are supported away from the section and from the fibres over a neighbourhood of infinity, so this lattice lies in Λ_n. Being unimodular of the same rank, it equals Λ_n. By Ebeling's theorem on complete vanishing lattices, as used in [Lön02, proof of Thm. 1], the reflection group has finite index in $\mathrm{O}(\Lambda_n)$.

Lönne's family is pulled back from V_n^{sm} by writing its branch data in Weierstrass form. More explicitly, writing the linear perturbation in his construction as $\ell(x,y) = ax + by$, the change of variables $X = -y$, $Y = z$ gives

$$Y^2 = X^3 + \varepsilon bX + x^{6n-1} - \varepsilon ax + u.$$

In homogeneous base coordinates $[s : t]$ with $x = t/s$, the corresponding Weierstrass coefficients are

$$a_4 = \varepsilon\, b\, s^{4n}, \qquad a_6 = t^{6n-1}s - \varepsilon a\, t s^{6n-1} + u s^{6n}.$$

These define the required map to the coefficient space; away from the surface discriminant, its image lies in V_n^{sm}. At $s = 0$, the coefficients satisfy $\mathrm{ord}_{s=0}(a_4) \geq 4n$ and $\mathrm{ord}_{s=0}(a_6) = 1$, so Tate's algorithm gives a fibre of type II [Tat75, §7, case (3)]; see also [BPS24, Thm. 1.6]. Thus the smooth members of Lönne's family lie in $V_n^{\mathrm{sm}} \smallsetminus T$, explaining why we pass through $\pi_1(V_n^{\mathrm{sm}})$. Since T is a non-empty Zariski-open subset of the smooth irreducible variety V_n^{sm}, the map

$$\pi_1(T) \longrightarrow \pi_1(V_n^{\mathrm{sm}})$$

is surjective. Hence the monodromy group over T contains a finite-index subgroup of $\mathrm{O}(\Lambda_n)$. Its intersection with the identity component of $\mathrm{SO}(\Lambda_{n,\mathbb{R}})$ is an arithmetic lattice by the Borel–Harish-Chandra theorem. Since this real semisimple group has no compact factors, the Borel density theorem shows that the Zariski closure contains $\mathrm{SO}(\Lambda_{n,\mathbb{Q}})$. Monodromy preserves the intersection form, so

$$H = \mathrm{SO}(\Lambda_{n,\mathbb{Q}}).$$

The variation $\mathbb{V}$ is a geometric polarizable integral variation of pure Hodge structure over the smooth connected quasi-projective variety T. André's normality theorem [And92, §5, Thm. 1] therefore gives $H \triangleleft G^{\mathrm{der}}$. Since G preserves the polarization up to scaling, its derived group preserves the intersection pairing and has determinant one. Consequently,

$$H \subseteq G^{\mathrm{der}} \subseteq \mathrm{SO}(\Lambda_{n,\mathbb{Q}}) = H.$$

Thus $G^{\mathrm{der}} = H$. For a connected reductive group, $G^{\mathrm{ad}} \simeq G^{\mathrm{der}}/Z(G^{\mathrm{der}})$. Hence $G^{\mathrm{ad}} \simeq \mathrm{PSO}(\Lambda_{n,\mathbb{Q}})$. An adjoint special orthogonal group in dimension at least five is simple over an algebraic closure, and therefore $\mathbb{Q}$-simple; here $\dim \Lambda_{n,\mathbb{Q}} = 12n - 4 \geq 20$. ∎

For the variation over T, a special subvariety is a closed irreducible algebraic subvariety maximal among those having the same generic Mumford–Tate group; the union of the proper special subvarieties is the strict Hodge locus [BKU24, §4.4]. The period dimension of a subvariety is the dimension of its image under the period map. A Mumford–Tate domain is a connected component of the orbit of a Hodge structure under the real points of its Mumford–Tate group. A special subvariety is atypical if its period dimension exceeds the dimension of the full period image minus the codimension of its Mumford–Tate domain in the generic domain. The definition also includes subvarieties whose period image is contained in the singular locus of the full period image [BKU24, Def. 2.2]. We use the dimension criterion below.

For $n \geq 9$, the next proposition places the positive-dimensional components of all simultaneous Hodge loci of 11 independent integral classes on marked deformation charts in one proper algebraic exceptional set. Outside that set, these fixed-class loci are zero-dimensional, which leads to countability.

Proposition 3.25. *Let $n \geq 9$. There is a proper Zariski-closed substack $E_n \subset \mathcal{W}_n^{\mathrm{sb}}$ such that, up to isomorphism, at most countably many surfaces in $\mathcal{W}_n^{\mathrm{sb}} \smallsetminus E_n$ have Mordell–Weil rank at least 11.*

Proof. Work on the coefficient presentation $T = V_n^{\mathrm{sb}}$ above, with primitive integral variation $\mathbb{V}$. By Lemma 3.24, its connected algebraic monodromy group is $\mathrm{SO}(\Lambda_{n,\mathbb{Q}})$, and its generic adjoint Mumford–Tate group is the $\mathbb{Q}$-simple group $\mathrm{PSO}(\Lambda_{n,\mathbb{Q}})$.

The Tate twist $\mathbb{V}(1)$ shifts each Hodge bidegree (p,q) to $(p-1,q-1)$, so primitive $(1,1)$-classes become classes of type $(0,0)$. The Tate twist changes neither monodromy nor the adjoint Mumford–Tate domain, and leaves the Hodge decomposition of the monodromy Lie algebra unchanged. In the remainder of this proof, special subvarieties and their Mumford–Tate groups are understood for $\mathbb{V}(1)$. In our simple orthogonal setting, Baldi–Klingler–Ullmo's level is the largest $|k|$ for which the monodromy Lie algebra has a non-zero $(k,-k)$-piece [BKU24, Defs. 4.13 and 4.15]. Such a piece consists of operators sending $H^{p,q}$ into $H^{p+k,q-k}$. The following calculation shows that the level is two. Via the polarization, the $(-2,2)$-piece of the orthogonal Lie algebra consists of skew maps from $H^{2,0}$ to $H^{0,2}$, giving

$$\mathfrak{so}(\Lambda_{n,\mathbb{C}})^{-2,2} \simeq \bigwedge^2 H^{0,2},$$

which is non-zero since $h^{0,2} = n-1 \geq 2$, while the pieces of degree $(k,-k)$ vanish for $|k| > 2$. Theorem 3.7 of Baldi–Klingler–Ullmo therefore shows that the union of the positive-period-dimensional atypical special subvarieties is algebraic [BKU24, Thm. 3.7]. This algebraic subset is contained in the strict Hodge locus, which is a countable union of proper closed algebraic subvarieties of T. By the Baire category theorem, this union has empty analytic interior, so the algebraic subset is proper. Its definition is invariant under changes of Weierstrass coordinates, so its reduced closed structure descends to a proper Zariski-closed substack $E_n \subset \mathcal{W}_n^{\mathrm{sb}}$. We call E_n the *exceptional Hodge locus*.

Let Φ denote the period map on a marked local deformation chart in $\mathcal{W}_n^{\mathrm{sb}}$. At its central point, the Schiffer formula and Lemma 3.4 give

$$d\Phi\left(\sum_{i=1}^{N} v_i \frac{\partial}{\partial t_i}\right)(\omega_0) = \sum_{i=1}^{N} v_i c_i(\omega_0)\eta_i.$$

Since the η_i form a basis and every $c_i(\omega_0)$ is non-zero, this differential is injective. Thus the period map is immersive on every marked local deformation chart. The coefficient space T has four additional parameters, corresponding to Weierstrass scaling and changes of coordinates on the base; the periods are locally constant along these directions. Its period image therefore has dimension N, which is the dimension used below in the expected intersection calculation.

Since the generic derived Mumford–Tate group is the full special orthogonal group, the generic Mumford–Tate domain is the orthogonal period-domain component considered in Subsection 2.2. Thus the codimension calculations from that subsection apply to the atypicality criterion.

Consider 11 independent integral classes which are simultaneously Hodge at a point of a marked chart. Their span is negative definite, and its fixed-class

period subdomain has codimension $11h$ by Subsection 2.2. Consider a positive-dimensional irreducible component of the simultaneous fixed-class Hodge locus. Algebraicity of Hodge loci [CDK95] realizes its lift to T locally as a branch of an algebraic Hodge-locus component. Enlarge that component to a special subvariety with the same generic Mumford–Tate group; see [BKU24, §4.4]. At a very general point of the local component, use the marking to regard this group as acting on $\Lambda_{n,\mathbb{Q}}$. It fixes the 11 classes in $\mathbb{V}(1)$, so its Mumford–Tate domain lies in their fixed-class subdomain and has codimension at least $11h$. Since

$$N - 11h = 9 - n \leq 0,$$

the expected dimension of its intersection with the period image is at most zero. The special subvariety contains the original component, which has positive period dimension, so its period dimension strictly exceeds this expected dimension. It is consequently atypical, and the original component lies in E_n.

It follows that, outside E_n, the simultaneous Hodge locus of every 11-tuple of linearly independent classes in Λ_n is zero-dimensional on each marked chart. There are countably many such integral 11-tuples, and every zero-dimensional analytic subset of a second-countable complex manifold is countable. Taking a countable cover by marked local deformation charts gives the assertion, using the integral Shioda identification on the all-I_1 locus. ■

Combining Proposition 3.25 with Corollary 3.13 and the section-moduli comparison of Proposition 3.15 gives the following rank-10 consequence.

Corollary 3.26. *Let $n \geq 9$. For every non-empty analytic open subset $U \subset \mathcal{W}_{\min,n}$ and every $H > 0$, there are a marked local deformation chart $B \to \mathcal{W}_n^{\mathrm{sb}}$ and a connected smooth 8-dimensional simultaneous fixed-class Hodge locus $Z \subset B$, with image in U, satisfying the following properties.*

1. *The family over Z carries 10 ordered independent sections whose Shioda classes are indivisible and flat. Their height-pairing matrix is constant on Z.*
2. *For every $b \in Z$ and every non-zero section $P \in \mathrm{MW}(S_b/\mathbb{P}^1)$, one has $\widehat{h}(P) > H$.*
3. *Outside an at most countable subset of Z, one has*
$$\mathrm{rk}\,\mathrm{MW}(S_b/\mathbb{P}^1) = 10.$$
4. *Forgetting the 10 chosen sections identifies the corresponding section-moduli germ with the germ of Z.*

Proof. Let E_n be the exceptional Hodge locus constructed in the proof of Proposition 3.25. Since $\mathcal{W}_n^{\mathrm{sb}}$ is irreducible and analytically dense in $\mathcal{W}_{\min,n}$, the intersection

$$U \cap (\mathcal{W}_n^{\mathrm{sb}} \smallsetminus E_n)$$

is non-empty and analytically open in $\mathcal{W}_{\min,n}$. Apply Corollary 3.13 there with $r = 10$. It gives a transverse simultaneous Hodge germ with dimension

$$10n - 2 - 10(n-1) = 8.$$

Choose a sufficiently small marked local quotient chart B at its base point and a connected smooth representative Z whose image remains in $U \cap (\mathcal{W}_n^{\mathrm{sb}} \smallsetminus E_n)$ and on which the bound in (2) holds.

Proposition 3.15 identifies the fixed-class Hodge germ with the section-moduli germ. After shrinking Z, the 10 chosen sections therefore extend over Z. Their Shioda classes remain flat, independent, and indivisible, and their height-pairing matrix is constant because the intersection pairing is flat.

Every surface parametrized by Z has rank at least 10. Proposition 3.25 gives at most countably many isomorphism classes of rank at least 11 outside E_n. Since $\mathcal{W}_{\min,n}$ is a separated Deligne–Mumford stack [PS25, Thm. 1.2(b)], a sufficiently small local quotient chart has finite fibres over isomorphism classes. Hence the corresponding set of points of Z is at most countable. ∎

Remark 3.27. The exceptional subset in Corollary 3.26 is at most countable. The proof does not show that it is discrete or rule out its being dense in Z. The argument gives no additional upper bound on the Mordell–Weil ranks at its points.

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